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Multimodal cell foundation models require synergistic integration beyond alignment

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Integrating measurements across modalities and scales into unified quantitative frameworks of cellular function is a central goal in computational biology. Although powerful unimodal foundation models exist, the scarcity of large-scale paired data limits joint multimodal training. This constraint favours compositional foundation models, which fuse frozen unimodal experts through a learned interface. However, it remains unclear when such fusion provides task-relevant information beyond the strongest unimodal representation rather than merely aggregating redundant signal. Here we introduce Synergistic Information Score, a metric grounded in partial information decomposition that quantifies information gain arising only from cross-modal interactions. Extending theory from self-supervised learning, we show that standard alignment-based objectives with frozen encoders collapse to detect linear redundancies, which constrains their ability to capture nonlinear synergistic states. This distinction is especially important for tasks that connect tissue morphology with gene expression. Benchmarking ten fusion methods on spatial transcriptomics datasets, we demonstrate that tasks dominated by linear redundancies are well served by unimodal baselines, whereas complex niche definitions benefit from synergy-aware integration. Finally, we perform a scaling analysis and show that fine-tuning a dominant unimodal expert is the most sample-efficient strategy for standard tasks. Together, these results suggest that progress in multimodal modelling of cellular state requires shifting from alignment-based objectives to synergy-driven integration that exploits complementary cross-modal signal.

Advances in self-supervised learning and the increasing scale of biological datasets have produced powerful encoders for individual modalities, including transcriptomics[1–4], histopathology[5–7] and spatial transcriptomics[8–10]. Recent multimodal efforts extend these representations to chromatin accessibility[11,12], proteomics[13] and text-based biological metadata[14–17]. However, multimodal data remain fundamentally asymmetric: millions of unimodal samples exist in isolation, whereas paired measurements capturing multiple modalities in the same cell or tissue context are much less available. This asymmetry favours compositional foundation models (CFMs), which reuse robust unimodal encoders as frozen experts and combine their outputs through learned interfaces[18]. These interfaces enable scalable multimodal modelling without requiring end-to-end retraining and have become central to efforts aimed at constructing integrated representations of cellular identity and function across biological scales[19–21] (see discussion on related work in Supplementary Information section 1).

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Still, a fundamental question remains: When does multimodal fusion truly add information beyond the strongest unimodal expert, and when does it merely repackage redundant signal?

Standard alignment-based fusion interfaces, which learn mappings between frozen unimodal representations into a shared embedding space (see the Methods for a formal definition), are optimized to maximize agreement between modalities. This is typically done by bringing representations of paired samples closer while separating unrelated samples through contrastive or regression-based objectives[22,23]. These objectives are effective for retrieval and matching tasks, where the goal is to collapse modalities onto a shared structure (for example, CLIP[22]). However, we argue that many scientific objectives require the opposite regime where discovery arises from synthesizing non-overlapping sets of information. For example, defining spatially varying cell states, niches or microenvironmental interactions may benefit from fusing tissue morphology and gene expression (GEX). In these regimes, we hypothesize that alignment-based fusion may reach a spectral ceiling defined here as a limit imposed by the eigenspectrum of pretrained representations, such that alignment recovers dominant linear correlations present in pretrained encoders but may fail to access synergistic information that emerges only through joint nonlinear processing. We investigate this limitation both theoretically and empirically.

In this work, we formalize and assess the limits of alignment-based CFMs under a central practical constraint of multimodal biology: unimodal datasets are abundant, whereas high-quality paired measurements remain comparatively scarce[19,21]. At the same time, integrating complementary modalities is increasingly recognized as essential for capturing cellular state, motivating accelerated investment in multimodal data generation and modelling[9,18,24]. As a result, many fusion strategies must operate in low-pairing regimes, where interfaces are trained with limited paired supervision. We demonstrate that simple alignment can appear effective while still primarily propagating redundant structure and suppressing modality-specific signal—a failure mode that is difficult to detect from downstream performance alone without an explicit diagnostic.

Our study can be broken down into three main components:

(1) First, we introduce the Synergistic Information Score (SIS), a diagnostic grounded in partial information decomposition (PID)[25,26], which quantifies whether a downstream task truly benefits from multimodal integration beyond the strongest unimodal expert. SIS measures the relative performance gain a fused representation provides over the best unimodal representation under a fixed probe class (here, linear probes), and thus, it captures when fusion makes additional task-relevant information accessible rather than merely aggregating redundant signal. In other words, SIS serves as a diagnostic to determine whether a task is unimodal-sufficient or cross-modal-dependent (Fig. 1).

(2) Second, we use SIS to show that many frozen-encoder alignment objectives saturate because they learn linear redundancy (that is, hit a spectral ceiling), which causes them to fail on cross-modal-dependent tasks. To explain this behaviour, we take spectral theory from the self-supervised learning literature and expand it to the multimodal frozen-encoder setting[23,27]. Our extension shows why diverse alignment objectives converge to equivalent spectral solutions that optimally recover redundancy but are limited in their ability to access nonlinear synergistic information.

(3) Third, we evaluate these predictions using a comprehensive experimental suite across multiple spatial transcriptomics datasets. Specifically, our evaluation uses a pulmonary fibrosis lung dataset[28,29], a developmental human thymus dataset[30,31] and a breast cancer dataset[32,33] (see Supplementary Information section 2 for dataset details). These datasets span regimes ranging from tight correspondence between modalities to substantial resolution mismatch. SIS separates regimes where unimodal fine-tuning suffices to extract all task-relevant information accessible to linear probes from regimes where multimodal integration provides additional predictive signal, exposing the value of paired data when modelling biology.

Beyond standard downstream tasks such as niche classification and cell-type composition regression, we evaluate spatial tasks that probe spatial understanding, including niche consistency, spatial consistency and neighbour prediction. These tasks span both unimodal-sufficient regimes, where transcriptomic information alone supports accurate prediction, and cross-modal-dependent regimes, where integrating morphology improves performance. We further perform a fine-tuning scaling analysis to characterize how the value of multimodal integration changes with dataset size. Across our benchmarks, synergy-aware objectives (for example, Contrastive MultiModal (CoMM)[34]) can break the spectral ceiling in cross-modal-dependent regimes, whereas alignment-based methods typically saturate near unimodal performance.

Together, our analyses establish SIS as a diagnostic of the information regime that governs if and how a given task benefits from multimodal learning. We show that when tasks are unimodal-sufficient, multimodal fusion provides limited additive benefit beyond strengthening the dominant modality; however, when tasks are cross-modal-dependent, integration can expose complementary signal that is inaccessible to unimodal representations alone. This distinction serves as a principled guide to whether improvements in predictive performance reflect unimodal refinement or genuine benefit from cross-modal pairing. More broadly, it also clarifies when multimodal foundation models can support truly synergistic inference—moving beyond correspondence to biological synthesis across scales.

Results

Multimodal integration in compositional cell foundation models

The scarcity of paired data limits end-to-end training of large multimodal models. This scarcity favours CFMs: architectures that combine robust, pretrained unimodal experts via a lightweight learned fusion interface. In this framework, the foundation model encoders are frozen, and the multimodal synthesis is entirely based on the fusion mechanism. To systematically evaluate these mechanisms, we established a controlled benchmarking framework (Fig. 1).

Our experimental design (Fig. 1a) rigorously isolates the contribution of the fusion strategy. Specifically, we extract spatially matched patches from whole-slide histopathology and spatial GEX data (Supplementary Information section 2), ensuring that both modalities view the same physical tissue region. We use state-of-the-art frozen foundation model encoders, using UNI-2[5] for histopathology and Nicheformer[8] for spatial transcriptomics, to transform matched patches from the same tissue slide into latent embeddings (Methods; Supplementary Information section 3). Because the foundation model encoders remain fixed, any difference in downstream performance is attributable solely to the fusion interface. We benchmark ten distinct fusion interfaces, ranging from simple aggregation (for example, concatenation (Concat)) to spectral (for example, canonical correlation analysis (CCA) and variance–invariance–covariance regularization (VICReg)[35]) and non-spectral objectives (for example, CoMM[34]) (Fig. 1b and Supplementary Information sections 4 and 5).

Standard benchmarks often obscure the distinction between simple redundancy and genuine multimodal integration. To resolve this, we evaluate unimodal and fused representations across five biologically grounded tasks that span local phenotyping, spatial organization and representation consistency (Fig. 1c). Niche classification and cell-type composition regression test whether representations capture local tissue identity and cellular composition. Spatial neighbourhood

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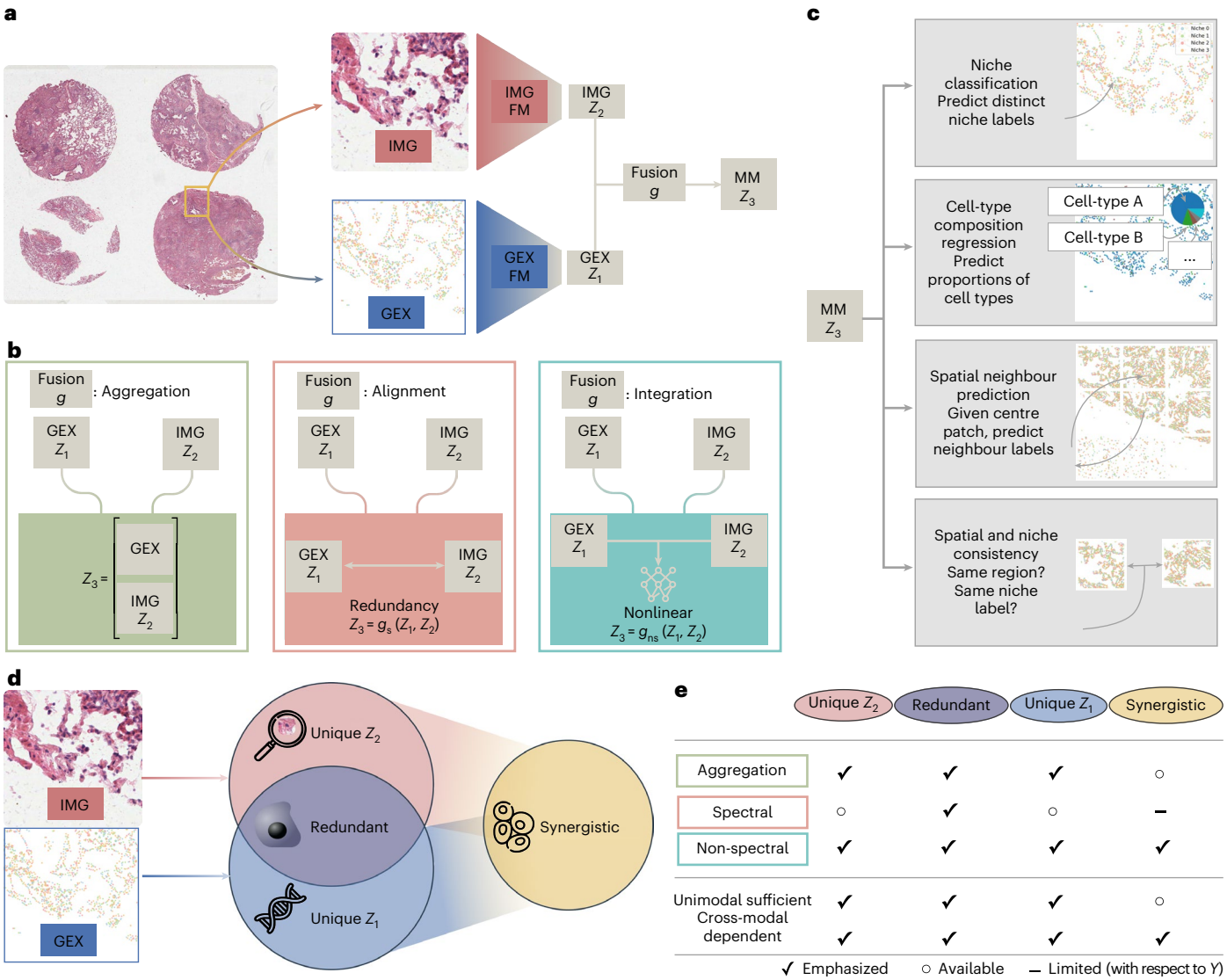


Fig. 1 | Fusion in CFMs. a, A schematic of the system architecture. We freeze unimodal foundation models (for histopathology images and gene expression, IMG and GEX, respectively) and train only the fusion interfaces g_1 and g_2 (summarized as g; see Methods for details); downstream evaluation uses the multimodal (MM) Concat of fused representations (Z_3) and a linear probe. This isolates the fusion mechanism as the sole variable contributing to the evaluation. **b**, A schematic of fusion methods: aggregation (left), spectral (s; middle) and non-spectral (ns; right). **c**, Downstream evaluation tasks probing different information regimes. We evaluate fused embeddings on niche classification, cell-type composition regression, spatial neighbourhood prediction and spatial and niche consistency. **d**, Conceptual decomposition of multimodal biological information into unique, redundant and synergistic components with respect to a downstream target Y (represented abstractly by the prediction tasks in **c**). Unique components are modality specific, redundant components are shared across modalities and synergistic components arise only through joint reasoning over modalities. **e**, Information-theoretic view of multimodal fusion. Different fusion strategies emphasize different information components. Symbols indicate whether unique, redundant or synergistic information (with respect to the target Y) is emphasized (✓), available (∘) or limited (–). **e** highlights which information components are emphasized, available or limited for each fusion strategy and downstream task. Credit: icons in **b** and **d**, Flaticon.com.

prediction asks whether a representation can infer broader tissue context beyond the local patch. And, finally, spatial consistency and niche consistency assess whether nearby patches or patches from the same niche are represented coherently. Together, these tasks distinguish unimodal-sufficient regimes, where one modality already captures the target-relevant information, from cross-modal-dependent regimes, where multimodal integration provides additional predictive signal.

The SIS distinguishes integration from alignment

A core question in evaluating CFMs is whether multimodal predictions benefit from genuinely complementary information or merely from aggregating redundant features across views. Standard downstream metrics such as F1 scores or R^2 quantify predictive accuracy but are agnostic to the information structure underlying that accuracy, specifically in distinguishing whether improvements stem from shared, unique or synergistic information between modalities. This distinction is central in biological settings, where cells are measured through multiple complementary views (for example, morphology and GEX) and where differences between measurements may either be noise to suppress or signal to exploit (Fig. 1d).

To make this distinction explicit, we introduce SIS, a metric grounded in PID that quantifies target-specific multimodal complementarity as the gain about a downstream variable Y obtained from the joint use of both modalities beyond the strongest unimodal source (Methods). As used here, SIS is an empirical diagnostic of accessible synergy under a specified downstream probe, and it formalizes the

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BOX 1

Using the SIS in practice

SIS as a diagnostic: SIS quantifies the performance gain of a fused representation relative to the best unimodal representation under a fixed probe class (here, linear probes). SIS distinguishes whether a task is unimodal-sufficient (reflected by SIS$\lesssim$0), where one modality already captures the relevant signal, or cross-modal-dependent (SIS>0), where integrating modalities makes additional task-relevant information accessible.

Recommended workflow:

(1) Evaluate on paired data: on a representative paired subset, train unimodal probe(s) and a fused probe, and compute SIS.
(2) Align modelling and data strategy with the information regime: for unimodal-sufficient tasks (SIS$\lesssim$0), improvements typically follow from strengthening the dominant modality. For example, this could be through higher-quality unimodal data, improved labels or targeted fine-tuning. For cross-modal-dependent tasks (SIS>0), multimodal integration can expose complementary signal not available in isolation, motivating investment in paired measurements and integration-aware interfaces. This regime is desired in synthesis-oriented objectives, such as efforts combining modalities into a unified representation of cellular state.
(3) Probe robustness: recompute SIS across task variants that modulate ambiguity or context (for example, increasing spatial distance or neighbourhood size). Increasing SIS indicates growing reliance on cross-modal interaction.

Interpretation: SIS is probe-dependent and, in this work, measures linearly accessible synergy. It should be interpreted as a diagnostic of the task's information regime rather than a prescriptive criterion: low SIS indicates that fusion does not improve accessibility of task-relevant signal under the current setup, whereas positive SIS identifies regimes where integration provides measurable benefit.

conceptual dichotomy illustrated in Fig. 1: alignment treats two measurements of the same entity as partially redundant views whose differences should be ignored, whereas integration treats those differences as informative and potentially interacting.

Formally, let Z_1 and Z_2 denote frozen unimodal representations, and let $Z_3 = [g_1(Z_1); g_2(Z_2)]$ be their fused representation produced by a fusion interface $\mathcal{G} = (g_1, g_2)$. Given a downstream task variable Y (for example, niche labels), we use the minimum mutual information (MMI) assumption[25,26] to define SIS as the relative information gain beyond the strongest unimodal baseline

$$\mathrm{SIS}(Y; Z_1, Z_2) = \frac{I(Y; Z_3) - \max(I(Y; Z_1), I(Y; Z_2))}{\max(I(Y; Z_1), I(Y; Z_2))}. \quad (1)$$

In practice, the intractable mutual information $I(Y;Z)$ is estimated via linear probes trained on the frozen representations (that is, Z_1 and Z_2 in the unimodal case, and Z_3 in the multimodal case). The empirical score $\widehat{\mathrm{SIS}}$ therefore measures probe-accessible multimodal gain over the strongest unimodal representation, rather than a unique PID decomposition. For classification tasks we use F1 Macro scores, and for regression tasks we use R^2. Restricting our evaluation to linear probes is crucial: SIS measures whether a fusion interface has rendered synergistic information linearly accessible, rather than deferring integration to complex downstream nonlinear models. Positive SIS indicates accessible synergy; near-zero or negative SIS indicates redundancy dominance or fusion failure. Under strong modality imbalance, SIS is intentionally conservative: fusion must improve over the dominant unimodal baseline to be positive, so low SIS indicates no additional target-relevant gain over that baseline, not that the weaker modality is biologically uninformative. Box 1 summarizes a practical workflow and links SIS to tasks that are unimodal sufficient versus those that are cross-modal dependent (Fig. 1e).

Evaluating ten fusion strategies across three spatial transcriptomic datasets (a pulmonary fibrosis lung atlas[28,29], a developmental human thymus dataset[30,31] and a breast cancer dataset[32,33]; Extended Data Figs. 1–3 and Supplementary Table 1) reveals that multimodal fusion is inherently task-dependent (Table 1). We next asked when multimodal fusion is helpful in the frozen-encoder setting and when a single modality suffices. To this end, SIS serves as a diagnostic of whether a given biological task benefits from multimodal learning or whether performance is already saturated by a unimodal expert.

Redundancy-dominated tasks constrain multimodal gains. Task dependence is evident in the breast dataset, which consists of large tissue patches with dense cellular content. Here, niche classification yields only small positive SIS values across the leading fusion methods (Table 1). CCA and CoMM achieve the largest gains (SIS of 0.058 and 0.045, respectively), while Concat remains close to the strongest unimodal baseline (SIS of 0.011). This places breast niche classification in a redundancy-dominated regime with bounded multimodal gains: fusion can provide modest additional signal, but the relevant information is already largely accessible from individual modalities or simple aggregation. Similar behaviour is observed in lung niche classification, where leading fusion methods achieve comparable SIS values. Together, these results show that multimodal fusion is not uniformly beneficial; its value depends on whether the task exposes complementary information beyond the strongest unimodal representation.

Synergy is required to integrate modalities. Aggregation fails when a task depends on integrating complementary information across modalities. In the thymus dataset, GEX is measured using Visium spatial transcriptomics[30,36], which captures transcripts over coarse spots spanning multiple cells and therefore does not align precisely with the fine-grained structure visible in histology images. This known resolution mismatch creates an integration regime where complementary spatial information is distributed across modalities. In this setting, multimodal methods achieve consistently positive SIS for the niche classification task, with the largest gains from non-spectral integration methods such as deep CCA (DCCA) and CoMM (SIS of 0.142 and 0.120, respectively; Table 1). Concatenation also improves over the strongest unimodal baseline but yields a smaller synergistic score (SIS of 0.072), while the spectral CCA baseline remains closer to the low-gain regime (SIS of 0.057). This pattern supports the PID interpretation: resolution mismatch induces a cross-modal-dependent task in which complementary information is distributed across modalities and is better exposed by learned integration than by aggregation or linear alignment alone. We further ablate the CoMM baseline[34], one of the strongest integration interfaces in our benchmark, and find that its performance depends on a combination of fusion, readout and contrastive-objective design, rather than any single component (Supplementary Tables 2 and 3).

These results demonstrate that, when correspondence across views is imperfect, effective fusion requires learning nonlinear cross-modal interactions rather than enforcing alignment. The same

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Table 1 | Task-dependent multimodal synergy across spatial transcriptomic datasets

		Lung (71,307)				Thymus (44,880)		Breast (126,764)	
		Niche classification		Cell-type composition		Niche classification		Niche classification	
Category	Method	F1 (↑)	SIS_{F1} (↑)	R^2 (↑)	SIS_{R^2} (↑)	F1 (↑)	SIS_{F1} (↑)	F1 (↑)	SIS_{F1} (↑)
Base	Random	0.046 ± 0.002	–	−0.064 ± 0.004	–	0.097 ± 0.002	–	0.070 ± 0.001	–
	IMG	0.631 ± 0.028	–	0.431 ± 0.004	–	0.332 ± 0.005	–	0.704 ± 0.009	–
	GEX	0.673 ± 0.017	–	**0.684 ± 0.007**	–	0.281 ± 0.006	–	0.706 ± 0.008	–
	Concat	0.713 ± 0.015	0.060	0.673 ± 0.007	**−0.017**	0.356 ± 0.008	0.072	0.714 ± 0.005	0.011
n-s	CoMM	**0.718 ± 0.023**	**0.067**	0.665 ± 0.010	−0.028	0.372 ± 0.003	0.120	0.738 ± 0.006	0.045
	BYOL	0.716 ± 0.024	0.064	0.649 ± 0.005	−0.052	0.353 ± 0.002	0.063	0.691 ± 0.004	−0.021
	SimSiam	0.501 ± 0.026	−0.255	0.608 ± 0.002	−0.112	0.349 ± 0.003	0.051	0.567 ± 0.023	−0.197
	DIM	0.703 ± 0.022	0.045	0.639 ± 0.012	−0.066	0.352 ± 0.007	0.062	0.692 ± 0.010	−0.020
	DCCA	0.627 ± 0.017	−0.068	0.642 ± 0.004	−0.063	**0.379 ± 0.004**	**0.142**	0.706 ± 0.006	0.000
a-s	SimCLR	0.715 ± 0.019	0.062	0.633 ± 0.005	−0.075	0.362 ± 0.003	0.090	0.712 ± 0.008	0.008
s	BT	0.658 ± 0.048	−0.022	0.606 ± 0.010	−0.115	0.354 ± 0.003	0.067	0.693 ± 0.010	−0.018
	CCA	0.714 ± 0.035	0.061	0.659 ± 0.004	−0.038	0.351 ± 0.004	0.057	**0.747 ± 0.008**	**0.058**
	VICReg	0.622 ± 0.031	−0.076	0.598 ± 0.014	−0.126	0.327 ± 0.005	−0.015	0.635 ± 0.010	−0.101

The columns report the niche classification F1 scores and SIS based on F1 (denoted as SIS_{F1}) for the lung, thymus and breast datasets. For the lung dataset, we also report the R^2 results for the cell-type composition regression task and the corresponding SIS based on R^2 (denoted as SIS_{R^2}). Note that SIS_{F1} and SIS_{R^2} are task-specific and reported per downstream metric. The values are the mean ± standard deviation across seeds; the dashes indicate unavailable metrics. The best-performing method for each task is listed in **bold**, and the second best method is underlined. Numbers in parentheses (71,307, 44,880 and 126,764) indicate the total number of spatially matched patches used for evaluation (see Supplementary Table 1 for details). Method categories include: 'base' (unimodal or trivial baselines), 's' (spectral alignment methods reducing to linear alignment), 'a-s' (approximately spectral methods converging to the spectral solution in limiting regimes) and 'n-s' (non-spectral integration methods introducing learned nonlinear interactions). Definitions and justification are given in Methods.

task yields weaker SIS in lung and breast (Table 1), where Xenium provides higher-resolution measurements and tighter local correspondence between morphology and GEX. In these regimes, aggregation and alignment perform closer to integration methods, indicating reduced, though not eliminated, dependence on cross-modal interaction. We report bootstrap confidence intervals and paired Wilcoxon signed-rank tests for the main evaluations in Supplementary Tables 4 and 5; these analyses support the regime-level SIS interpretation while indicating that fine rank differences among close fusion methods are not individually significant under the fivefold evaluation.

Unimodal dominance renders integration unnecessary. The lung cell-type composition task provides a matched control regime: unlike expert-defined niche labels, which reflect histological tissue regions and architecture, cell-type composition is a broad cellular-fraction that is estimated from Xenium cell annotations and is therefore closely aligned with local GEX. GEX is a primary molecular determinant of cell identity, as established by large-scale single-cell atlases and transcriptomic profiling studies[37,38]. As a result, for this downstream task, we reasoned that histology would provide information that is redundant to what is strongly conveyed through GEX. Consistent with this, SIS is negative across all fusion methods (for example, CoMM scored SIS of −0.028), indicating that multimodal fusion does not provide additional task-relevant information beyond the strongest unimodal representation, which in this task is GEX (see Table 1, where GEX R^2 = 0.684 is the highest among the unimodal baselines). This demonstrates the diagnostic value of SIS: it identifies regimes where integration is unnecessary because the relevant signal is already accessible through a single modality.

SIS regimes are robust to encoder choice and probe capacity. We evaluated SIS under shallow and deep multilayer perceptron (MLP) decoders as a probe-capacity control. The strongest and most consistent increases in synergy occurred for thymus niche classification, whereas unimodal-sufficient regimes remained low-SIS and lung niche classification exhibited probe-dependent instability partly driven by reduced unimodal performance (Extended Data Fig. 4 and Supplementary Information section 6). Next, we repeated the SIS analysis over the encoder grid available in the original HFD feature release[29,31,33], spanning three image encoders and four GEX representations. Across 12 encoder pairs per task, the qualitative regime structure is preserved (Extended Data Fig. 5 and Supplementary Information section 6): thymus remains the clearest high-SIS regime, lung and breast niche classification show smaller gains, and lung cell-type composition remains close to the unimodal-sufficient boundary. Fine rankings vary across encoder pairs, so this serves as a robustness control for SIS regimes rather than a benchmark of external encoders.

From alignment to integration. Together, these results establish a PID-based view of multimodal fusion (Fig. 1d,e). Fusion interfaces applied to frozen unimodal encoders do not uniformly improve performance but instead reflect the task's underlying information regime. Spectral alignment methods preferentially extract shared structure but may suppress modality-specific signal, aggregation methods preserve modality-specific information but do not model interactions and non-spectral integration methods enable access to cross-modal synergy. Consequently, the design of multimodal cell foundation models should be guided not by a single universal fusion strategy but by the information structure induced by the biological question and downstream task of interest.

Synergistic integration recovers spatial tissue organization

We next asked how the cross-modal-dependent regime evolves as tasks require increasing spatial context. Spatial context is a defining property of biological tissues: cellular function depends not only on local composition but also on structured interactions across space. Multimodal cell foundation models must therefore support representations that capture tissue-wide organization rather than memorizing local correlations. This capability is reflected in whether models extrapolate to regions physically beyond immediate observation. We hypothesized that, as prediction distance increases and local ambiguity grows, performance should increasingly depend on integrating complementary information

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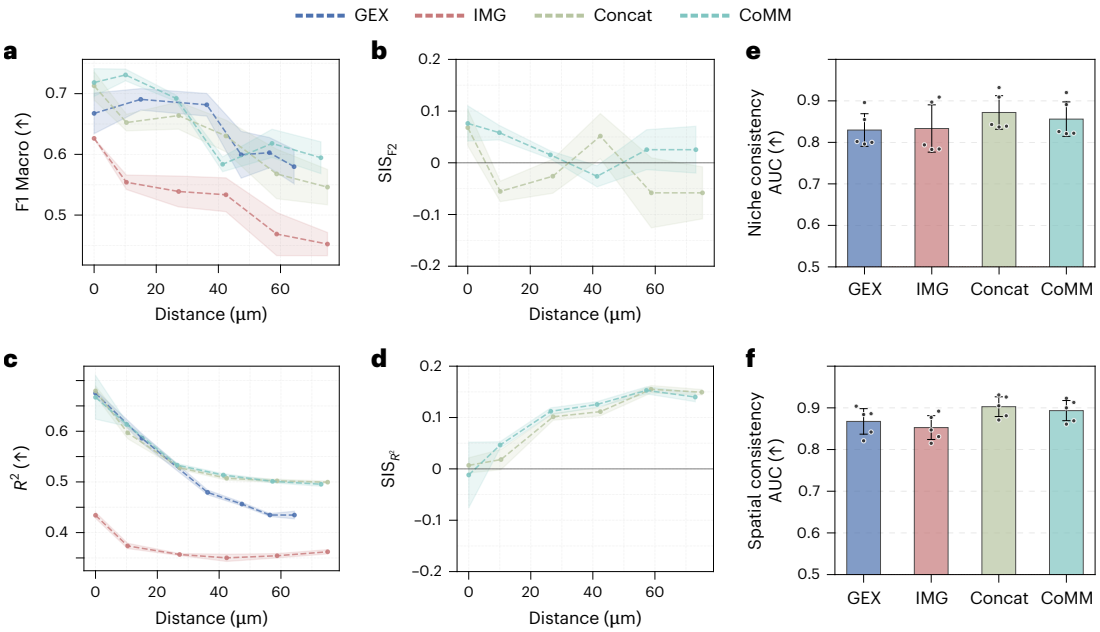


Fig. 2 | Benchmarking frozen encoders and their fusion on spatial neighbourhood tasks in the lung dataset. Spatial neighbour prediction on the lung dataset where performance and corresponding SIS are plotted as a function of the distance between a centre patch and a neighbour patch. **a**,**b**, Neighbour niche classification with F1 Macro (**a**) and with SIS_{F1} (**b**) is shown. **c**,**d**, Neighbour cell-type composition regression with R^2 (**c**) and with SIS_{R^2} (**d**) is shown. SIS quantifies the performance gain of a fused representation relative to the strongest unimodal representation; the horizontal line marks SIS of 0. In **a** and **c**, we show unimodal GEX (blue) and imaging (IMG; red) alongside multimodal fusion methods Concat (green) and CoMM (turquoise). In **b** and **d**, we report SIS only for the multimodal fusion methods (Concat, CoMM), because SIS is defined relative to the best unimodal baseline. **e**,**f**, Niche consistency (**e**) and spatial consistency (**f**) on the lung dataset, quantified as AUC for a consistency classifier computed using the similarities between embeddings (Methods). Shaded regions and error bars denote mean ± 1 s.d. across five cross-validation folds. Distances with insufficient valid neighbour pairs for a given modality are omitted.

across modalities. We probe this capability by measuring SIS in a spatial neighbourhood prediction task (Fig. 2a–d), where frozen unimodal encoders and their multimodal fusion are evaluated on predicting niche identity and cell-type composition of neighbouring patches at increasing spatial distances (that is, going from the centre to a neighbour patch).

This setting explicitly stress-tests fusion objectives (Fig. 2a–d). At short centre-to-neighbour distances, unimodal representations already perform strongly and SIS remains near zero, indicating unimodal-sufficient regimes. As this spatial distance increases, unimodal performance degrades more rapidly than multimodal fusion and SIS correspondingly rises (Fig. 2d), indicating a transition towards a cross-modal-dependent regime where integration becomes increasingly beneficial.

Multimodal fusion preserves tissue consistency. We first verify that fusion does not distort the intrinsic biological structure encoded by unimodal experts. To do this, we quantify niche and spatial consistency via pairwise probes that predict (from two patch embeddings) whether patches share the same niche label or spatial region, summarized by area under the curve (AUC; Methods). Both unimodal encoders exhibit strong spatial and niche consistency (AUC >0.8; Fig. 2e,f, Extended Data Fig. 6 and Supplementary Information section 7), reflecting the strong spatial priors learned during pretraining. Importantly, multimodal fusion preserves this structure across all datasets (Extended Data Fig. 6). Multimodal fusion models achieve consistency comparable to, or marginally exceeding, unimodal baselines, indicating that fusion establishes a shared latent space without collapsing or corrupting the spatial manifold. We additionally verify that fusion interfaces preserve spatial and semantic neighbourhood structure via pairwise consistency diagnostics (Supplementary Tables 6 and 7 and Supplementary Information section 9). This confirms that subsequent gains in spatial prediction arise from information synthesis rather than representational degradation.

Synergy emerges as spatial context expands. Predicting properties of neighbouring patches reveals a clear transition from redundancy-dominated to integration-dominated regimes (Fig. 2a–d). At short centre-to-neighbour distances (≤10 μm), the neighbour patch remains within the local microenvironment of the centre patch, and unimodal signals are relatively predictive (Fig. 2a,c). In this regime, unimodal encoders perform strongly, and multimodal fusion offers only limited improvements over aggregation (Fig. 2a,c), consistent with alignment being sufficient. As spatial distance increases (20–60 μm), local correspondence weakens, and unimodal performance degrades more rapidly than multimodal performance. Notably, in neighbour cell-type composition regression, a continuous task that depends on tissue-scale context rather than discrete local cues, degradation is driven primarily by the GEX unimodal baseline, while imaging remains comparatively stable (Fig. 2c; some distances are unavailable for specific modalities due to dataset filtering and variable neighbour availability). Accordingly, SIS increases sharply with distance for this task (from ~0 to >0.15; Fig. 2d), whereas SIS for neighbour niche classification remains comparatively flat (Fig. 2c). This pattern indicates that as spatial distance grows, predictive success increasingly depends on integrating complementary signals across modalities rather than exploiting shared redundancy.

Convergence towards integration under spatial ambiguity. Comparing datasets further highlights how known differences in spatial resolution modulate the utility of fusion (Supplementary Information sections 7 and 8). The lung and breast datasets utilize Xenium in situ technology, where GEX and tissue morphology are measured at comparable effective resolution and exhibit strong local correspondence, resulting in largely redundant representations at short range. Consistent with this, SIS is low or negative at short range, and simple Concat performs competitively (Fig. 2b,d and Extended Data Fig. 6a–d). By contrast, the thymus dataset uses Visium spatial transcriptomics, which

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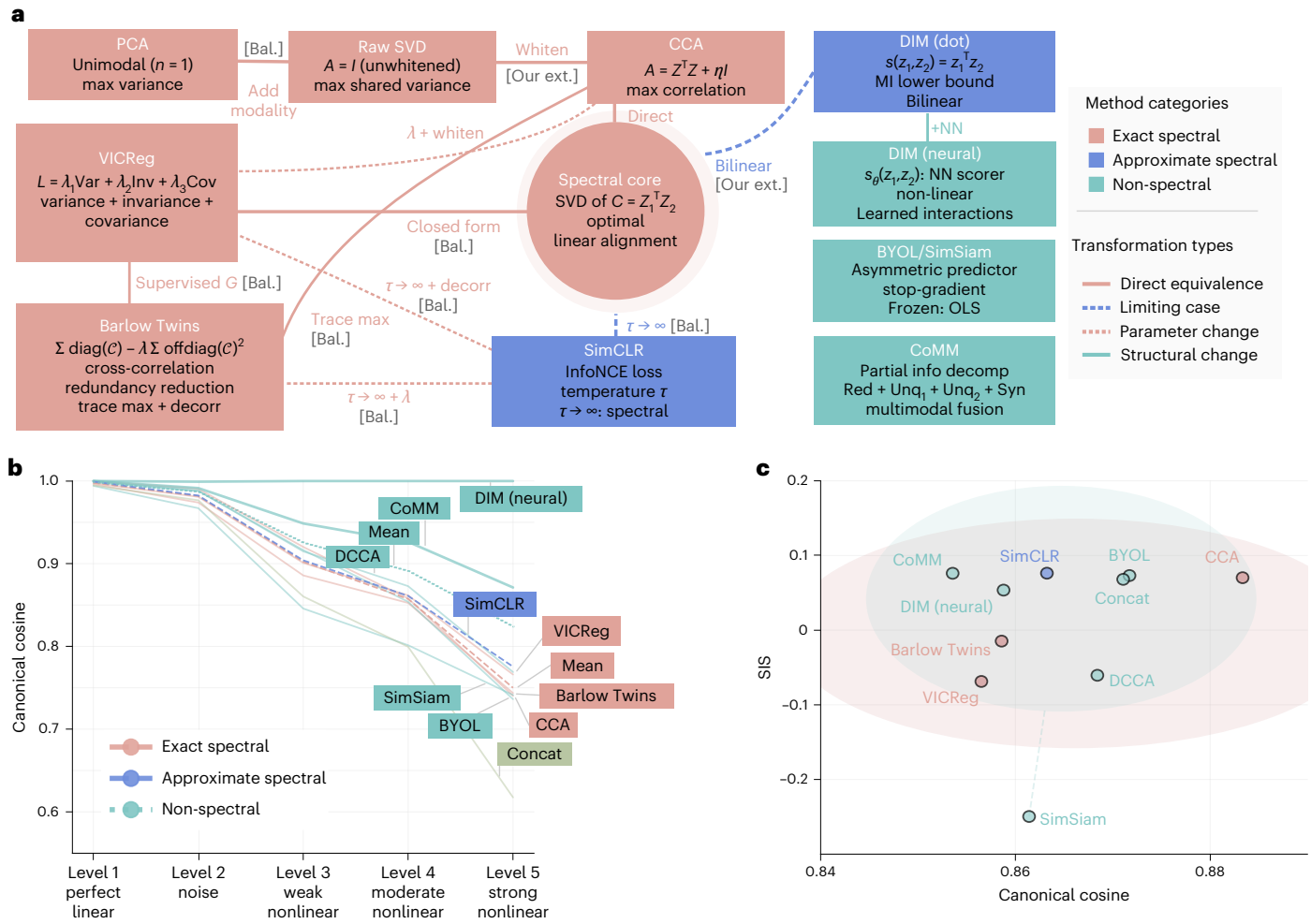


Fig. 3 | Assessing the spectral ceiling of the multimodal frozen-encoder framework. a, Theoretical taxonomy of multimodal fusion objectives under frozen encoders. Exact spectral alignment methods (salmon: principal component analysis (PCA), raw SVD, CCA, VICReg, Barlow Twins) reduce to closed-form SVD/CCA solutions of the cross-covariance matrix. Approximate spectral methods (purple: SimCLR, Deep InfoMax (DIM) dot) converge to the same solution in limiting regimes. Non-spectral integration methods (teal: DIM neural, BYOL, Simple Siamese (SimSiam), CoMM) introduce asymmetric predictors or synergy-aware terms that break the eigenvalue structure and permit nonlinear interactions. [Bal.] denotes connections made in ref. 27, and [Our ext.] denotes connections made in our extension of their analysis. **b**, Synthetic evaluation under increasing nonlinearity (Methods). Canonical cosine measures similarity between the recovered shared signal and the ground-truth shared structure in the generator (higher indicates closer recovery). Solid lines show individual methods, dashed lines show category means and shaded regions represent 95% confidence intervals. **c**, Alignment-synergy landscape on the lung dataset. Each point is a fusion method, plotted by the canonical cosine to the spectral (linear) solution (x axis; higher means closer agreement with the SVD solution) and SIS (y axis). This plot illustrates the empirical relationship between the proximity to the linear alignment solution and the measured multimodal gain under SIS for this dataset. n, number of modalities; Z_1, Z_2, frozen unimodal representations; $\tilde{Z}_1, \tilde{Z}_2$, projected representations (outputs of the fusion heads g_1, g_2); $C = Z_1^\top Z_2$, cross-covariance of the frozen representations (scaled by $1/N$ in Methods); A, constraint matrix in the trace maximization $W_j^\top A_j W_j = I$ (Methods), where $A = I$ gives the unwhitened SVD and $A_j = Z_j^\top Z_j + \eta I$ gives CCA; I, identity; η, ridge regularization hyperparameter; L, loss; $\lambda_1, \lambda_2, \lambda_3$, weights of the VICReg variance (Var), invariance (Inv) and covariance (Cov) terms; $\mathcal{C}$, cross-correlation matrix of $\tilde{Z}_1$ and $\tilde{Z}_2$; diag/offdiag, diagonal and off-diagonal entries of $\mathcal{C}$; λ (Barlow Twins), weight of the off-diagonal term; decorr, decorrelation (penalty on off-diagonal entries); G, positive-pair relation matrix of ref. 27 (supervised G: pairs defined by class labels; distinct from the fusion interface $\mathcal{G}$); z_1, z_2, projected representations of one paired sample (rows of $\tilde{Z}_1, \tilde{Z}_2$); s, critic score; s_θ, neural critic with parameters θ; MI, mutual information; NN, neural network; OLS, ordinary least squares; InfoNCE, noise-contrastive estimation loss; τ, temperature; Red, Unq1, Unq2, Syn, redundant, unique (modality 1, 2) and synergistic information about the target Y (equation (7)).

captures GEX over coarse spots spanning multiple cells and therefore introduces inherent spatial ambiguity relative to high-resolution histology. In this regime, SIS is already positive at short distances for neighbour niche classification (Extended Data Fig. 6f), and the synergy-aware CoMM model consistently outperforms aggregation. A class-wise thymus case study shows that these gains are niche-specific, with CoMM improving selected capsular and corticomedullary compartments rather than uniformly improving all labels (Extended Data Fig. 7 and Supplementary Information section 8). Despite these differing starting points, SIS converges to a positive value across all datasets at long range (SIS of ~0.03–0.07), indicating a shared setting in which spatial context dominates local redundancy and learned integration becomes essential regardless of initial measurement resolution.

When spatial integration becomes necessary. Together, these results show that, for frozen-encoder fusion, the benefit of multimodal integration in neighbourhood prediction grows with spatial range and is amplified under resolution mismatch. In regimes with strong local correspondence, fusion provides limited gains over unimodal baselines and simple aggregation, whereas in regimes with

increased spatial ambiguity, integration-aware interfaces yield higher SIS and better preserve performance at larger distances (Fig. 2c,d and Extended Data Fig. 6). This suggests that modelling the emergent structure of tissue biology requires moving beyond objectives that merely align views towards interfaces that actively synthesize complementary spatial information.

A spectral ceiling constrains frozen alignment

The observation that many fusion methods converge to similar performance plateaus (Table 1) points to a shared theoretical constraint inherent to the frozen-encoder setting. Building on self-supervised learning theory showing that diverse unimodal objectives reduce to spectral solutions[23,27], we extend this analysis to multimodal fusion. Here, the spectral ceiling refers to the frozen-encoder CFM regime in which linear fusion maps with whitening or variance constraints reduce alignment to an eigen- or singular-value problem on the cross-covariance between modalities. In this regime, the learned interface is restricted to recovering dominant linear shared structure, providing a reference model for when alignment emphasizes redundancy rather than task-relevant cross-modal interactions. This extension reveals a fundamental distinction: alignment objectives optimize linear redundancy under a spectral ceiling, whereas integration objectives can, in principle, capture nonlinear synergy beyond this limit.

Frozen encoders reduce alignment to linear redundancy. In the frozen-encoder regime, we show that a broad class of fusion objectives reduces, under linear alignment mappings and standard variance or whitening constraints, to maximizing the trace $\mathrm{Tr}(C)$, where $C = \tilde{Z}_1^\top \tilde{Z}_2/N$ denotes the cross-covariance between aligned representations $\tilde{Z}_1 = g_1(Z_1)$ and $\tilde{Z}_2 = g_2(Z_2)$ for N observations. By the Eckart–Young theorem[39], the optimal solution is given by the singular value decomposition (SVD) of C. In the Methods, we formalize this reduction by showing that, under frozen encoders and linear fusion, a wide class of objectives is equivalent to maximizing $\mathrm{Tr}(C)$ up to normalization constraints (that is, whitening during training and L_2-normalization at evaluation). Crucially, the variance or whitening constraints imposed by these objectives are necessary to exclude trivial collapsed solutions (for example, $\tilde{Z}_1 = 0$), but they also enforce a linear spectral structure. As a result, such methods act as alignment interfaces: they optimally recover linear correlations shared between modalities but are theoretically restricted to information expressible within the linear subspace of $(\tilde{Z}_1, \tilde{Z}_2)$. By construction, they cannot exploit the task-relevant structure that appears only through nonlinear cross-modal interactions. By contrast, integration approaches (for example, CoMM and Bootstrap Your Own Latent (BYOL)) introduce asymmetry or synergy-aware terms that prevent reduction to this eigenvalue problem, allowing them to escape the spectral ceiling (Fig. 3a).

Non-spectral methods are robust to nonlinearity. We validate this distinction using synthetic data with controlled nonlinear coupling between modalities (Methods). In this simulation study, canonical cosine is computed against the generator's ground-truth shared signal and therefore measures recovery of the underlying cross-modal relationship (Fig. 3b). Although all methods recover the signal in linear regimes, spectral methods degrade rapidly as nonlinearity increases; whereas non-spectral methods degrade at a slower rate, maintaining substantially higher canonical cosine under strong nonlinearity. These synthetic experiments serve as controlled analogues of increasing modality mismatch. The five levels are fixed generative scenarios that progressively stress linear spectral recovery. This confirms that spectral alignment is inherently limited to linear structure, whereas non-spectral objectives can capture nonlinear cross-modal interactions. As an additional boundary analysis, we show that partial or full encoder training changes the geometry of cross-modal representation, confirming that the spectral ceiling is an isolated result for frozen compositional encoders rather than a global characterization of end-to-end multimodal optimization (Extended Data Fig. 8 and Supplementary Information section 10).

In real biological data, ground truth is unavailable. Here, canonical cosine instead quantifies similarity to the spectral reference solution (that is, SVD/CCA; Fig. 3c). Under this interpretation, proximity to the spectral solution reflects recovery of linear redundancy, while deviations indicate representations that move beyond purely linear alignment. Consistent with the synthetic results, methods that remain close to the spectral solution do not necessarily achieve high SIS, whereas several non-spectral methods achieve higher SIS despite deviating from the linear optimum (Extended Data Fig. 9). Across datasets, higher SIS is often achieved by methods that depart from this spectral reference, consistent with integration gains arising under biological mismatch or resolution effects that are not fully captured by linear alignment. Non-spectral methods deviate from the linear optimum to capture complementary structure, achieving higher SIS across datasets (Extended Data Fig. 9 and Supplementary Section 11).

Scaling analysis reveals that unimodal fine-tuning is more sample-efficient than alignment

A central practical question for deploying cell foundation models is how to allocate limited labelled data: Should one invest in adapting a strong unimodal expert, or in fusing multiple modalities through a multimodal interface? This tradeoff is especially relevant given the distinct information regimes identified in Fig. 1e, which distinguish tasks that are unimodal-sufficient from those that are genuinely cross-modal dependent. To resolve this empirically, we perform the following scaling analysis. First, we progressively fine-tune the GEX encoder on increasing fractions of downstream data (1% to 100%), while keeping the image encoder frozen, thereby creating a controlled setting to compare unimodal adaptation against frozen multimodal fusion. We then evaluate whether multimodal fusion provides additive value over unimodal adaptation (Fig. 4 and Supplementary Information sections 12 and 13). A complementary alignment and data scaling analysis, where we vary the paired CoMM training set from 1% to 100%, is reported in Supplementary Table 8. This analysis directly investigates whether performance gains arise primarily from strengthening the dominant modality or from integrating complementary signal across modalities.

Unimodal fine-tuning is highly sample-efficient. In the lung niche classification task, fine-tuning the GEX encoder yields rapid gains at small data fractions (Fig. 4a), with similar qualitative trends also seen in the breast and thymus datasets (Extended Data Fig. 10). Moving from frozen features (0%, where only the downstream probe is trained) to 1% fine-tuning substantially improves F1 Macro, after which performance largely saturates with additional data (Fig. 4a). Across all training fractions, the fine-tuned GEX encoder matches or exceeds the multimodal baselines (Concat and CoMM), indicating that adapting the strongest modality captures most attainable performance. A similar pattern holds for cell-type composition regression, where the fine-tuned GEX encoder remains the strongest performer, while multimodal fusion yields only modest improvements relative to simple aggregation and does not close the gap to the unimodal baseline (Fig. 4b). Overall, these scaling curves indicate that, for this dataset and set of tasks, most attainable performance is captured by adapting the strongest unimodal expert, consistent with a unimodal-sufficient regime (Fig. 1e).

Fine-tuning reduces the incremental value of fusion under SIS. We next track SIS as the GEX expert is adapted (Fig. 4c,d). Once GEX is fine-tuned (from 1% onward), SIS is consistently negative for both tasks, indicating that fused representations do not outperform the best unimodal representation under the linear-probe evaluation used to define SIS. This behaviour reflects a redundancy-dominated regime in which unimodal adaptation increasingly captures the accessible task-relevant

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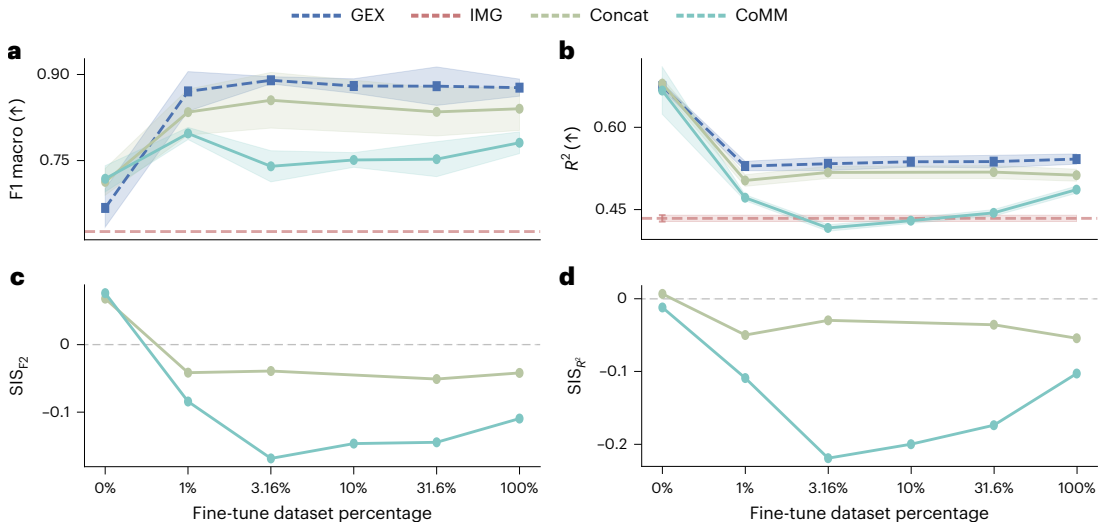


Fig. 4 | Scaling behaviour of unimodal fine-tuning and multimodal fusion. **a**,**b**, Performance on the lung dataset as a function of labelled downstream training data used to fine-tune the GEX encoder (x axis; depicted as log-scaled fractions); we report F1 Macro for niche classification (**a**) and R^2 for cell-type composition regression (**b**). Here 0% indicates frozen encoders with only a linear probe trained, and x% indicates fine-tuning GEX on that fraction while keeping IMG frozen. **c**,**d**, Corresponding SIS values computed from the same probe metrics, showing SIS_{F1} (**c**) and SIS_{R2} (**d**). Curves show unimodal GEX and IMG baselines and two fusion interfaces (Concat and CoMM). Shaded regions denote the mean ± 1 s.d. across five cross-validation folds.

signal, leaving little (linearly accessible) complementary information to be unlocked through integration. Under strong modality imbalance, negative SIS should be interpreted as evidence that fusion does not improve over the dominant expert under the evaluated readout, not as evidence that the second modality contains no biological information. Accordingly, when the downstream task is dominated by information already present in a single modality, fine-tuning that expert is the most sample-efficient strategy. In such unimodal-sufficient regimes (Fig. 1e), frozen multimodal alignment introduces additional parameters without unlocking new task-relevant information.

Resolution mismatch preserves the value of integration. The scaling behaviour on the thymus and breast datasets follows the information regimes identified in earlier sections (Extended Data Fig. 10 and Supplementary Sections 12 and 13). In the thymus dataset, where GEX is measured at coarse Visium resolution and modalities provide complementary information, multimodal integration remains competitive across all fine-tuning fractions. Although unimodal GEX performance improves gradually with fine-tuning, it does not clearly subsume the multimodal models, and the synergy-aware CoMM interface maintains stable performance and consistently positive SIS throughout. This identifies the thymus task as cross-modal-dependent, demonstrating that SIS correctly diagnoses a regime in which integration methods provide additive value beyond unimodal adaptation. Accordingly, under resolution mismatch and spatial ambiguity, complementary information from a second modality remains accessible even as the dominant expert is refined. By contrast, the breast dataset exhibits behaviour more similar to the lung dataset, with high baseline performance and limited gains from doing either fine-tuning or fusion, indicating restricted space for additional improvement.

Scaling clarifies efficiency regimes for multimodal foundation models. Together, these results establish that SIS provides a principled diagnostic for determining whether unimodal adaptation suffices or if multimodal integration provides additive value. In the lung dataset (and similarly in breast), fine-tuning GEX rapidly saturates performance and drives SIS to become negative, indicating limited multimodal benefit under this setup and therefore a unimodal-sufficient regime (Fig. 4 and Extended Data Fig. 10a–d). In the thymus dataset, where resolution mismatch induces cross-modal dependence, CoMM maintains positive SIS across training fractions, indicating a cross-modal-dependent regime (Extended Data Fig. 10e–h). In other words, when tasks are unimodal-sufficient, adapting the dominant expert provides the most sample-efficient path to optimal performance; however, when tasks are cross-modal-dependent, multimodal integration can expose complementary information that remains inaccessible to unimodal refinement alone. Thus, multimodal integration is not a universal performance amplifier, but a mechanism whose benefit depends on whether the downstream task fundamentally requires combining information distributed across modalities.

Discussion

Constructing computational models that predict cellular function across biological scales requires more than aggregating multimodal measurements. It requires biological synthesis: combining complementary views to resolve ambiguity that cannot be resolved within any single modality. Yet, as unimodal datasets continue to grow and multimodal acquisition efforts begin to expand, current evaluation paradigms often conflate improvements from recovering redundant cross-modal structure with genuine integration. This conflation creates a practical blind spot: simple fusion objectives may appear effective even when they primarily propagate the strongest unimodal signal. To expose and investigate this conflation, we introduce the SIS as a diagnostic for multimodal learning. By formalizing the distinction between alignment (recovering shared structure) and integration (making complementary information jointly accessible), SIS reveals that strong benchmark performance can arise even when a model effectively relies on a single dominant modality. This diagnostic perspective complements recent work applying PID to characterize multimodal datasets and models more broadly[40]; our focus is specifically on whether fusion interfaces make synergistic information linearly accessible in the frozen-encoder setting, and we aim to describe the spectral mechanism that explains when and why they fail to do so.

Our theoretical analysis identifies a mechanism that explains this pattern in a widely used practical workflow, where frozen backbones with a learned fusion interface are evaluated by linear probes. In this setting, frozen encoders fix the geometry of each modality, and many standard fusion objectives (including CCA-like or contrastive formulations

under variance/whitening constraints) reduce to maximizing linear cross-covariance. This 'spectral ceiling' is well suited to retrieval-focused objectives where modalities are expected to match in a shared latent space, but it also implies that fusion interfaces are biased towards extracting the linear redundancy already present in pretrained experts. As a consequence, these interfaces reliably recover shared signals while offering limited capacity to expose synergistic interactions that are not linearly expressible in the aligned features. Importantly, this ceiling is specific to the frozen-encoder regime and to the common practice of assessing cross-modal alignment primarily with linear metrics. In end-to-end CLIP-style training, deep encoders can map complex correspondences into representations that appear approximately linearly aligned, although nonlinear modality gaps can persist in practice. Together, these considerations motivate objectives that go beyond geometric alignment and explicitly target information gain, such as CoMM.

Empirically, the alignment versus integration distinction translates into a concrete design principle for multimodal model design. In redundancy-dominated regimes with high cellular resolution and precise cross-modal correspondence (for example, Xenium-like spatial profiling), unimodal experts already capture most task-relevant signal; correspondingly, SIS is near zero or negative. Here, our scaling analysis shows that unimodal fine-tuning is the most sample-efficient path to strong performance, and multimodal fusion provides little consistent improvement. By contrast, the promise of multimodal modelling lies in regimes where ambiguity or resolution mismatch distributes task-relevant information across modalities. For example, in spot-based spatial transcriptomics, coarse GEX measurements and fine-grained histology create an integration regime in which simple aggregation can fail to make complementary signal accessible; similarly, predicting tissue context at an increasing spatial range shifts tasks away from local redundancy towards cross-modal dependence. This resolution contrast explains why Visium-based niche classification in the thymus exhibits higher SIS than the Xenium-based niche tasks in the lung and breast, where molecular and morphological signals are more locally matched. In these settings, alignment alone is insufficient, and synergy-aware integration becomes competitive. Methods such as CoMM model cross-modal interactions rather than only enforcing agreement, enabling access to structure that is not recoverable from either modality alone under a linear probe.

These findings establish two information regimes that govern multimodal learning (Fig. 1e and Box 1), with SIS serving as a diagnostic to distinguish between them. In unimodal-sufficient regimes (SIS $\lesssim 0$), task-relevant signal is already accessible within a single modality, and improving the strongest unimodal foundation model provides the most efficient path to performance gains. In cross-modal-dependent regimes (SIS >0), complementary information becomes accessible through fusion, and multimodal fusion interfaces that model cross-modal interactions provide additive value. Although this study focuses on spatial histology applications, evaluating SIS in other multi-omic settings remains an important direction for future work. For example, as a pilot experiment, we explored the use of SIS on multi-omics data using GEX paired with either measurements of surface protein abundance (for example, antibody-derived tags) or chromatin accessibility assays[41,42] (Supplementary Table 9 and Supplementary Information section 14). More broadly, SIS provides a modality-agnostic diagnostic for model design: it identifies when unimodal refinement or simple aggregation suffices, and when integration-oriented fusion is required to expose task-specific information beyond the strongest unimodal baseline. This distinction is particularly relevant for synthesis-oriented settings, where the objective is to combine information across biological scales rather than to recover redundant structure, aiming for regimes where fusion yields positive SIS (that is, where cross-modal interactions become measurably accessible).

Ultimately, our work argues for a shift in multimodal learning from enforcing correspondence to achieving biological synthesis. Models that synthesize information across biological scales cannot be built by merely forcing modalities to correlate; they require objectives that preserve each modality's unique structure while making emergent properties accessible through interaction. We introduce SIS as a diagnostic probe for information regimes in multimodal learning and demonstrate that synergy-aware, integration objectives such as CoMM can outperform alignment objectives in cross-modal-dependent settings. Together, this contribution provides practical guidance for when to prioritize unimodal refinement versus multimodal integration when holistically modelling cellular state.

Methods

SIS

In this section, we describe the overall problem statement that motivates the SIS. Given paired raw input data matrices from two modalities $X_1 = [\mathbf{x}_1^{(1)}, \ldots, \mathbf{x}_1^{(N)}]$ and $X_2 = [\mathbf{x}_2^{(1)}, \ldots, \mathbf{x}_2^{(N)}]$ (for example, histology images and GEX profiles), together with a downstream task measured by the output vector $\mathbf{y} = [y^{(1)}, \ldots, y^{(N)}]$ (for example, niche labels or cell-type proportions), we define frozen unimodal representations as the following

$$Z_1 = f_1(X_1), \qquad Z_2 = f_2(X_2), \tag{2}$$

where f_1 and f_2 are frozen unimodal encoders and $Z_i \in \mathbb{R}^{N \times D_i}$ with D_i being the dimension of the embedded space. We next introduce the concept of a fusion interface, denoted by $\mathcal{G} = (g_1, g_2)$, which operates on the frozen unimodal representations. Here, $g_1 : \mathbb{R}^{D_1} \to \mathbb{R}^{K_1}$ and $g_2 : \mathbb{R}^{D_2} \to \mathbb{R}^{K_2}$ are modality-specific fusion mappings (also referred to as projection heads), yielding

$$\tilde{Z}_1 = g_1(Z_1), \qquad \tilde{Z}_2 = g_2(Z_2), \tag{3}$$

where $\tilde{Z}_i \in \mathbb{R}^{N \times K_i}$ with K_i denoting the dimension of the modality-specific fusion. Depending on the objective, the fusion interface may instantiate an alignment-based or an integration-based fusion strategy. All fusion objectives considered in this work learn projection mappings g_1 and g_2 from unimodal representations (Z_1, Z_2), with optimization objectives defined on the resulting projected representations $(\tilde{Z}_1, \tilde{Z}_2)$. In the spectral analysis, we explicitly restrict to the linear regime where $g_i(Z_i) = Z_i W_i$, which enables closed-form eigenvector or SVD. Outside this regime, g_i may be nonlinear, but the spectral characterizations no longer apply. We then form a single fused representation

$$Z_3 := [\tilde{Z}_1; \tilde{Z}_2] \in \mathbb{R}^{N \times (K_1 + K_2)}, \tag{4}$$

and train a linear probe h to predict the output vector $\mathbf{y}$ for downstream evaluations. As part of our motivation for SIS, the goal is to develop a scalar measure that has the following properties:

- It quantifies the complementarity between Z_1 and Z_2 with respect to a downstream task;
- It is grounded in information theory yet tractable in high-dimensional representation spaces;
- It enables task-dependent comparison across multimodal fusion methods.

Below, we review key information-theoretic preliminaries before deriving the SIS.

Mutual information. The mutual information between a target random variable $Y \in \mathcal{Y}$ (where the output vector $\mathbf{y}$ contains realizations of Y) and a source random variable $Z \in \mathcal{Z}$ is defined as

$$I(Y; Z) = H(Y) - H(Y|Z), \tag{5}$$

where $H(Y) = -\sum_y p(y) \log p(y)$ denotes the entropy of Y, and $H(Y|Z)$ denotes the conditional entropy. By the data processing inequality[43],

for any (possibly randomized) decoding function $d : Z \to \hat{Y}$, we have $I(Y; d(Z)) \leq I(Y; Z)$.

PID. PID[25,26] decomposes the mutual information between a target variable Y and two source variables into redundant, unique and synergistic components. Reference 40 recently demonstrated the utility of scalable PID estimators for characterizing such interactions across general multimodal benchmarks. In our setting, the sources are the projected representations $\tilde{Z}_1 = g_1(Z_1)$ and $\tilde{Z}_2 = g_2(Z_2)$ produced by the fusion interface, and we define the joint representation as their concatenation $Z_3 := [\tilde{Z}_1; \tilde{Z}_2]$. As Concat is a bijective reparameterization of the pair $(\tilde{Z}_1, \tilde{Z}_2)$, mutual information is invariant under this transformation, and we have

$$I(Y; Z_3) = I(Y; \tilde{Z}_1, \tilde{Z}_2). \tag{6}$$

PID therefore yields

$$I(Y; Z_3) = \mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2) + \mathrm{Unq}(Y; \tilde{Z}_1) + \mathrm{Unq}(Y; \tilde{Z}_2) + \mathrm{Syn}(Y; \tilde{Z}_1, \tilde{Z}_2). \tag{7}$$

Here, the redundant component $\mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2)$ quantifies information in Y that can be explained by either projected representation alone, the unique components $\mathrm{Unq}(Y; \tilde{Z}_1)$ and $\mathrm{Unq}(Y; \tilde{Z}_2)$ quantify information accessible exclusively from one projected representation, and the synergistic component $\mathrm{Syn}(Y; \tilde{Z}_1, \tilde{Z}_2)$ quantifies information in Y that is only accessible when both projected representations are considered jointly.

Deriving the SIS. From equation (7) and by definition of PID, the mutual information between Y and a single source consists of the information it shares with the other source (redundancy) and the information that is unique to itself[25,26]. Equivalently, synergy contributes only to the joint information $I(Y; \tilde{Z}_1, \tilde{Z}_2)$ and not to $I(Y; \tilde{Z}_i)$ for either source individually

$$I(Y; \tilde{Z}_1) = \mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2) + \mathrm{Unq}(Y; \tilde{Z}_1), \tag{8}$$

$$I(Y; \tilde{Z}_2) = \mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2) + \mathrm{Unq}(Y; \tilde{Z}_2). \tag{9}$$

Solving for the synergistic component then yields

$$\mathrm{Syn}(Y; \tilde{Z}_1, \tilde{Z}_2) = I(Y; Z_3) - I(Y; \tilde{Z}_1) - I(Y; \tilde{Z}_2) + \mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2). \tag{10}$$

To obtain a tractable target-specific approximation of synergy, we instantiate the redundancy term with the MMI principle

$$\mathrm{Red}(Y; \tilde{Z}_1, \tilde{Z}_2) \approx \min(I(Y; \tilde{Z}_1), I(Y; \tilde{Z}_2)). \tag{11}$$

This choice treats the information about Y contained in the weaker source as being redundant with the stronger source, yielding the following

$$\mathrm{Syn}(Y; \tilde{Z}_1, \tilde{Z}_2) \approx I(Y; Z_3) - \max(I(Y; \tilde{Z}_1), I(Y; \tilde{Z}_2)). \tag{12}$$

MMI is an operational approximation rather than a unique definition of redundancy; note that alternative PID formalisms may assign different redundancy, unique-information and synergy components[44,45]. In practice, SIS compares the fused representation against the strongest original unimodal representation from either Z_1 or Z_2. Since each aligned representation $\tilde{Z}_i = g_i(Z_i)$ is a deterministic function of Z_i, the data processing inequality implies $I(Y; \tilde{Z}_i) \leq I(Y; Z_i)$ for $i \in \{1, 2\}$. Replacing $\max(I(Y; \tilde{Z}_1), I(Y; \tilde{Z}_2))$ by the stronger baseline $\max(I(Y; Z_1), I(Y; Z_2))$ therefore yields a conservative estimate of the fusion gain. We then define SIS as

$$\mathrm{SIS}(Y; Z_3) := \frac{I(Y; Z_3) - \max(I(Y; Z_1), I(Y; Z_2))}{\max(I(Y; Z_1), I(Y; Z_2))}. \tag{13}$$

Intuitively, SIS measures the relative amount of task-relevant information made accessible by fusion beyond the strongest unimodal baseline under the chosen probe class. Positive SIS indicates accessible synergy, whereas near-zero or negative values indicate redundancy dominance or fusion degradation, respectively.

Note on approximation error and interpretation. Let $I_{\max} = \max(I(Y; Z_1), I(Y; Z_2))$ such that equation (13) can be written as $\mathrm{SIS} = [I(Y; Z_3) - I_{\max}]/I_{\max}$. The MMI assumption in equation (12) is exact when the weaker modality contributes no unique information about Y beyond the stronger modality. Deviations arise when the weaker modality contains unique target-relevant information, in which case SIS under the MMI framework captures joint gain over the strongest unimodal source rather than just pure synergy. To formalize this, assume without loss of generality that $I(Y; Z_1) \geq I(Y; Z_2)$. Then

$$I(Y; Z_3) - I(Y; Z_1) = \mathrm{Syn}(Y; Z_1, Z_2) + \mathrm{Unq}(Y; Z_2 \setminus Z_1). \tag{14}$$

Thus, positive SIS can theoretically arise from synergistic information and from unique information contributed by the weaker modality. Values near zero indicate that fusion adds little target-relevant information beyond the best unimodal source, whereas negative values indicate degradation under the chosen representation and probe. In our biological setting, modalities are measured over the same tissue structure and therefore we expect substantial target-relevant redundancy. Nevertheless, in this work, we report SIS as a measure of joint gain.

Mathematical properties and important edge cases of SIS. The SIS satisfies the following mathematical properties:

- Symmetry: $\mathrm{SIS}(Y; Z_3) = \mathrm{SIS}(Y; Z_3^{\mathrm{swap}})$, where $Z_3 := [g_1(Z_1); g_2(Z_2)]$ and $Z_3^{\mathrm{swap}} := [g_2(Z_2); g_1(Z_1)]$. This holds because Z_3^{swap} differs from Z_3 only by a permutation of feature coordinates, and the linear probe hypothesis class used to estimate SIS is invariant under such permutations.
- Sign: SIS >0 indicates synergy, while SIS <0 indicates fusion degradation.
- Task-specificity: SIS depends on the target label Y, distinguishing tasks that truly require cross-modal reasoning.

In addition to these properties, there are a few edge cases that are worth noting:

- Redundant modalities: if $I(Y; Z_1) = I(Y; Z_2) = I(Y; Z_3)$, then SIS = 0.
- Fusion failure: if $I(Y; Z_3) < \max(I(Y; Z_1), I(Y; Z_2))$, then SIS <0, indicating that fusion degrades performance below a better unimodal baseline.

Computationally efficient estimation of SIS via linear probes. For high-dimensional representations, computing the exact mutual information between the task variable Y and representation variable Z is intractable. We therefore approximate mutual information using the empirical performance of a linear probe h trained on a given representation matrix Z, denoted as $\mathrm{Perf}(Z)$. In our setting, Z is instantiated either as a unimodal representation (Z_1 or Z_2) or as the multimodal representation $Z_3 := [\tilde{Z}_1; \tilde{Z}_2]$. We then compute SIS as the following:

$$\widehat{\mathrm{SIS}} = \frac{\mathrm{Perf}(Z_3) - \max(\mathrm{Perf}(Z_1), \mathrm{Perf}(Z_2))}{\max(\mathrm{Perf}(Z_1), \mathrm{Perf}(Z_2))}, \tag{15}$$

where performance is measured via a F1 Macro score for classification tasks, and performance is determined via R^2 for regression-based tasks. Thus, $\widehat{\mathrm{SIS}}$ is a performance-indexed approximation: it measures the target-relevant gain made accessible by the fused representation relative to the strongest unimodal representation under the same probe class. Although standard performance metrics (such as F1 scores) are

not strictly additive in the same way as entropy, $\widehat{\mathrm{SIS}}$ serves as a practical estimator of the relative predictive gain for several reasons:

- By the data processing inequality, any decoder $h : Z \to \hat{Y}$ can only extract information present in the representation, meaning, $I(Y; h(Z)) \leq I(Y; Z)$ (ref. 43). In our setting, h is a linear probe (classifier or regressor), so its performance reflects a lower bound on the task-relevant information in Z that is linearly accessible (also see discussion around equation (5)).
- In the context of foundation models, we care about accessible information that downstream tasks can readily extract, not just from an information-theoretic capacity.
- By restricting the probe to be linear, $\widehat{\mathrm{SIS}}$ measures whether the fusion interface $\mathcal{G} = (g_1, g_2)$ has rendered synergistic information linearly accessible, which is a useful metric for evaluating whether a foundation model provides ready-to-use representations. If synergy exists but is confined to a complex nonlinear manifold that a linear probe cannot access, then the interface has failed in its purpose.

Therefore, our proposed approach provides a conservative estimate of the amount of task-relevant synergistic information that is linearly accessible from the fused representation. If a linear probe cannot extract information, the fusion has failed to make synergistic patterns readily usable. Again, a positive SIS indicates the extraction of accessible synergistic states; whereas a near-zero or negative SIS implies redundancy or fusion failure, respectively. In the main text, we denote the empirical estimator $\widehat{\mathrm{SIS}}$ simply as SIS when no ambiguity arises and specify the probe-dependent metric via indices such as $\mathrm{SIS}_{\mathrm{F1}}$ and SIS_{R^2}.

Additional computational details. The empirical way of computing SIS is done post hoc using downstream performance metrics on held-out samples. Given the two unimodal representations and the fused representation, SIS only requires estimating the ratio in equation (15) and therefore adds negligible computational overhead. Its reliability is governed by the uncertainty of the underlying F1 or R^2 values rather than by a separately trained information estimator. As a result, we report bootstrap confidence intervals for SIS using the same folds as the downstream evaluations and paired Wilcoxon signed-rank tests with Holm correction within each dataset-task family (Supplementary Tables 4 and 5 and Supplementary Information section 7). The dominant costs are upstream: one-time encoder inference over N patches, optional fusion-interface training on cached embeddings, export of the representations and fitting the probe on K-folds downstream (Supplementary Table 10 and Supplementary Information section 15). In the frozen-encoder setting, the encoder inference is amortized across fusion methods and downstream tasks because embeddings are cached. Fusion-interface training scales with the number of paired patches and training epochs, while probe fitting scales with the number of examples, the latent representation dimension and the number of cross-validation folds.

Spectral alignment theory

We extend the spectral theory of self-supervised learning to the multimodal setting with frozen encoders. [27] showed that diverse unimodal self-supervised learning objectives (for example, VICReg, Barlow Twins and simple framework for contrastive learning (SimCLR)) admit spectral interpretations and derived closed-form optima in the linear regime under standard variance/whitening constraints. [23] further analysed a spectral contrastive loss and connected it to contrastive learning objectives from the augmentation graph perspective. These works motivate that, when representations are fixed, broad classes of self-supervised learning objectives reduce to constrained spectral optimization problems. As part of our contributions, we adapt this viewpoint to analyse multimodal alignment under frozen encoders, where the goal is to couple representations from two different fixed-modality encoders rather than augmented views of the same input.

Spectral problem setup. We adopt the same notation and fusion interface defined earlier in this section. Frozen encoders f_1 and f_2 produce representations $Z_1 \in \mathbb{R}^{N \times D_1}$ and $Z_2 \in \mathbb{R}^{N \times D_2}$. A fusion interface $\mathcal{G} = (g_1, g_2)$ maps these to projected representations $\tilde{Z}_1 = g_1(Z_1) \in \mathbb{R}^{N \times K_1}$ and $\tilde{Z}_1 = g_1(Z_1) \in \mathbb{R}^{N \times K_1}$. Here, we analyse fusion objectives that are defined on the paired projected representations $(\tilde{Z}_1, \tilde{Z}_2)$.

Spectral methods primer. Spectral methods analyse matrices through their eigenvalues and eigenvectors, which identify orthogonal directions capturing dominant variance and correlation structure (see ref. 46 for a textbook summary). For paired representations $Z_1 \in \mathbb{R}^{N \times D_1}$ and $Z_2 \in \mathbb{R}^{N \times D_2}$, their shared structure is characterized by the cross-covariance $C = Z_1^{\top} Z_2 / N$. The SVD of this matrix $C = U \Sigma V^{\top}$, yields orthogonal directions that maximize alignment between the two spaces, with singular values quantifying alignment strength. These directions arise as solutions to constrained alignment problems of the form $\max_{W_1, W_2} \mathrm{tr}(W_1^{\top} C_{12} W_2)$ subject to orthonormality constraints, which are solved by the top singular vectors (Eckart–Young theorem[39]). Whitening or variance constraints ensure non-degenerate solutions and isolate correlation structure. Thus, under linear projections and variance normalization, alignment objectives reduce to identifying the dominant singular vectors of cross-covariance matrices, which define the optimal shared subspace between representations.

The linear regime. To obtain closed-form spectral characterizations, we restrict to the linear regime in which the fusion mappings are linear maps applied row-wise

$$g_i(Z_i) = Z_i W_i, \qquad W_i \in \mathbb{R}^{D_i \times K_i}. \tag{16}$$

This restriction is used solely for theoretical purposes. Note that in the general setting, g_i may be nonlinear, in which case the spectral closed-form solutions derived below no longer apply. We do not assume that the relationship between modalities is linearly realizable in the frozen feature space; rather, the linear-regime analysis characterizes the optimizer of a broad class of alignment objectives under linear heads. Moreover, variance/whitening (or analogous) constraints are essential to exclude collapsed solutions (for example, $W_1 = W_2 = 0$) and yield non-trivial spectral structure. This distinction matters because:

- Frozen geometry: unlike end-to-end training, frozen encoders impose a fixed-representation geometry.
- Constrained spectral optima: under linear heads and standard variance/whitening constraints, several covariance-based objectives (for example, VICReg and Barlow Twins) admit spectral solutions[27]. Closely related spectral perspectives for contrastive objectives have also been studied[23].

The spectral ceiling. We define the spectral ceiling as the constrained optimum attained by redundancy-maximizing alignment objectives in the linear-head and frozen-encoder regime. Although practical implementations may use nonlinear heads, the linear derivation below establishes a reference bound for objectives whose solutions concentrate on cross-covariance structure. In the linear regime, a canonical constrained trace maximization problem is

$$\max_{W_1 \in \mathbb{R}^{D_1 \times K_1},\, W_2 \in \mathbb{R}^{D_2 \times K_2}} \mathrm{tr}\,(W_1^{\top} Z_1^{\top} Z_2 W_2) \text{ subject to } W_j^{\top} A_j W_j = I_{K_j}. \tag{17}$$

This problem is solved by the SVD of an appropriately whitened cross-covariance matrix (via the Eckart–Young theorem[39]). Here I_{K_j} denotes the $K_j \times K_j$ identity matrix. For our frozen-encoder setting, we state this classical result:

Proposition 1. (Spectral solution for frozen-encoder alignment) Assume Z_1, Z_2 are column-centred. The trace maximization problem in equation (17) with constraints $W_j^\top A_j W_j = I_{K_j}$, where K_j is the jth target dimension and $A_j \succcurlyeq 0$, is solved by the top $K = \min(K_1, K_2)$ singular vectors of the whitened cross-covariance

$$\tilde{C} = A_1^{-1/2}\left(\frac{1}{N} Z_1^\top Z_2\right) A_2^{-1/2}. \tag{18}$$

Let $\tilde{C} = \tilde{U}\Sigma\tilde{V}^\top$ be the SVD. Then one optimal choice is

$$W_1^* = A_1^{-1/2}\tilde{U}_{1:K}, \qquad W_2^* = A_2^{-1/2}\tilde{V}_{1:K}. \tag{19}$$

Proof. This follows directly from the Eckart–Young theorem[39]. The CCA case corresponds to $A_j = Z_j^\top Z_j/N + \eta I$ as established by Hotelling[47], with η denoting a regularization hyperparameter.

Note that this proposition characterizes the spectral optimum of linear alignment interfaces on fixed representations. It does not rule out nonlinear or end-to-end multimodal models that alter the representation geometry itself.

Categorization of fusion methods. In the main text, we categorize methods by whether their objectives reduce to equation (17) in the linear limit (see also Fig. 3a), following the spectral viewpoints in ref. 27 and related analyses of contrastive objectives in ref. 23. We refer to methods whose linear-limit objectives admit closed-form spectral solutions as spectral methods. We then further classify approaches whose linear-limit objectives reduce directly to the closed-form spectral solutions under standard variance/whitening (anti-collapse) constraints as exact spectral methods.

- Linear CCA[47]: directly solves a generalized eigenvalue problem.
- VICReg[35] and Barlow Twins[48]: with frozen representations and linear heads, these objectives admit constrained spectral solutions[27].

We describe approximately spectral methods as objectives whose optimization is closely tied to spectral structure in certain limiting regimes (for example, large-batch, many-negative settings for contrastive learning), while we avoid requiring an exact reduction for the specific loss used in practice.

- SimCLR (InfoNCE)[49]: a contrastive objective with close connections to spectral viewpoints under appropriate limits and assumptions (see, for example,[23] for a spectral contrastive formulation and related analysis).

Lastly, we describe non-spectral methods as objectives whose structure prevents reduction to a symmetric trace maximization eigenproblem. This class includes the following list.

- Naive Concat: Joins representations without learning g_1 and g_2. Since it does not perform redundancy reduction or subspace selection, it preserves all information (redundant and unique) and does not collapse to the spectral subspace. In the main text, we also refer to Concat as Aggregation, as it aggregates information from both modalities.
- DCCA: optimizes the canonical correlation objective using nonlinear projection heads. Unlike linear CCA, the nonlinear transformations allow it to capture correlations on a learned manifold, thereby escaping the linear cross-covariance spectrum of the frozen inputs.
- DIM (neural)[50]: maximizes mutual information using a parameterized neural critic (discriminator) rather than a simple dot-product. This neural estimation allows it to capture complex nonlinear dependencies that do not reduce to trace maximization.
- BYOL[51] and SimSiam[52]: uses asymmetric prediction with stop-gradient. With frozen encoders, this becomes an asymmetric regression problem rather than a symmetric trace maximization.
- CoMM[34]: uses a PID-aware objective with specific synergy terms that explicitly deviate from redundancy maximization.

Additional details for each method are provided in Supplementary Information section 4.

Empirical analysis of spectral convergence. To empirically test our categorization, we quantify how closely each method's learned projected embeddings $(\tilde{Z}_1, \tilde{Z}_2)$ adhere to the linear spectral solution, even when using nonlinear projection heads. We compute a reference linear solution via the SVD of the centred cross-covariance $C = Z_1^\top Z_2/N$ on $N = 1{,}000$ paired patches from the held-out test set. Let $C = U\Sigma V^\top$ and denote the top-K singular subspaces by $U_{1:K}$ and $V_{1:K}$. We use parallel analysis[53] to estimate an effective dimensionality K by permuting rows of Z_2 independently $L = 100\times$ to break pairing while preserving marginals, and taking the count of observed singular values exceeding the 95th percentile of the null singular values. Note that, in our experiments, this yielded $K = 5$ for which we verified stability across random seeds and train/test splits.

Convergence metrics. We evaluate adherence to the reference spectral subspaces using: (1) a Procrustes-style subspace distance, (2) projection energy onto the reference subspace and (3) canonical cosine or principal angle-based similarity. Exact definitions and implementation details are given in Supplementary Information section 11. A higher canonical cosine indicates closer adherence to the linear redundancy solution, whereas a lower canonical cosine indicates deviation from this linear subspace (which can reflect either nonlinear structure capture or failure, depending on downstream SIS).

Synthetic data generator. We designed a five-level synthetic data generator that progressively increases nonlinearity and noise, enabling controlled evaluation of spectral versus non-spectral methods. The rationale behind this design is to create a systematic progression from perfect linear alignment to strong nonlinear coupling, with each level characterized by intrinsic complexity measures rather than target canonical cosine values. All levels use a shared latent structure: we generate a shared latent $W \in \mathbb{R}^{N_{\text{syn}} \times K_{\text{syn}}}$ from a standard Gaussian, where $N_{\text{syn}} = 20{,}000$ and $K_{\text{syn}} = 10$. Both modalities are constructed via orthonormal projection bases $U \in \mathbb{R}^{D_1 \times K_{\text{syn}}}$ and $V \in \mathbb{R}^{D_2 \times K_{\text{syn}}}$ (with $D_1 = D_2 = 128$), scaled to achieve canonical correlations.

- Level 1 (perfectly linear): here, we generate a perfect linear relationship where $Z_1 = WU^\top$ and $Z_2 = WV^\top$, with no private noise (residual standard deviation $\sigma_1 = \sigma_2 = 0$). In other words, the signal-to-noise ratio is infinite. This provides an ideal baseline for spectral methods.
- Level 2 (noise degradation): this scenario has the same linear structure as level 1 but with increasing levels of private noise $\sigma_1, \sigma_2 \in \{0.0, 0.1, 0.2, 0.3, 0.5, 0.8, 1.0\}$ added independently to each modality: $Z_1 = WU^\top + \epsilon_1$ and $Z_2 = WV^\top + \epsilon_2$ where $\epsilon_j \sim \mathcal{N}(0, \sigma_j^2 I)$. This creates a systematic signal-to-noise ratio degradation going from infinite to finite values.
- Level 3 (linear with private structure): here, we generate a linear relationship with structured private noise (where standard deviations are set to $\sigma_1 = \sigma_2 = 0.3$). This represents the case where modalities share a common signal, but each also contains modality-specific information (signal-to-noise ratio of ~3.3).
- Level 4 (moderate nonlinear coupling): in this case, we apply a nonlinear transformation to the second modality such that

$Z_1 = WU^\top + \epsilon_1$ and $Z_2 = [\tanh(W) + 0.2\sin(W)]V^\top + \epsilon_2$ with $\sigma_1 = \sigma_2 = 0.3$. The combination of tanh saturation and sinusoidal modulation introduces moderate nonlinearity while preserving some monotonicity.

- Level 5 (strong nonlinear coupling with modality rotation): this last scenario has the strongest nonlinearity via a rotated nonlinear transformation to the second modality such that $Z_1 = WU^\top + \epsilon_1$ and $Z_2 = [\tanh(WR)]V^\top + \epsilon_2$, where $R \in \mathbb{R}^{K_{syn} \times K_{syn}}$ is a random rotation matrix. Here, we use increased asymmetric noise ($\sigma_1 = 0.5$, $\sigma_2 = 0.7$). This creates substantial nonlinearity (compositional structure: saturation ∘ rotation) and breaks linear alignment structure. The asymmetric noise ($\sigma_1 \neq \sigma_2$) tests robustness to modality imbalance.

These levels are fixed by the data-generating parameters, including private noise, structured private variation, transform class and nonlinearity strength. They are not optimized to induce specific canonical-cosine values; canonical cosine is measured only after training as an evaluation statistic. The analysis relies on the qualitative contrast between linear regimes and nonlinear regimes rather than on sharp boundaries between adjacent levels. This progression systematically tests method robustness: spectral methods should excel at the first two levels but degrade with increasing nonlinearity, while non-spectral methods should maintain higher canonical cosine across all levels. The resulting canonical cosine values are measured empirically, not engineered to specific targets. We extend the spectral theory of self-supervised learning to the multimodal setting with frozen encoders. Reference 27 showed that diverse unimodal self-supervised learning objectives (for example, VICReg, Barlow Twins and SimCLR) converge to spectral solutions recovering eigenmodes of kernel matrices. Reference 23 further showed that the spectral contrastive loss concentrates on the top eigenfunctions of the data distribution. These works established that many seemingly different self-supervised learning objectives reduce to trace maximization problems when data representations are fixed. As part of our contributions in this work, we adapt this framework to analyse multimodal alignment under frozen encoders. This is a distinct setting from unimodal self-supervised learning, as it involves fusing representations across two different fixed-modality encoders rather than augmented views of the same input.

Encoder-trainability boundary experiment

To test the boundary between frozen compositional fusion and end-to-end multimodal training, we repeated the synthetic spectral analysis under three encoder-trainability regimes: frozen encoders, last-layer trainable encoders and fully trainable encoders. For each regime, we trained either a spectral-alignment objective or a CoMM-style integration objective on the same five-level synthetic ladder used in Fig. 3, ranging from linear shared structure to strong nonlinear coupling. We report two quantities: (1) canonical cosine to the oracle shared generative directions, measuring recovery of the true shared signal, and (2) the relative change in cross-modal representation geometry from initialization. The latter quantifies how much the representation space on which the frozen-encoder spectral solution is defined changes during training. This experiment is not used to redefine the spectral ceiling; it instead tests when its fixed-representation assumptions cease to apply.

Downstream task evaluation on real data

We evaluate unimodal and fused representations across five biologically grounded tasks with increasing spatial complexity. All downstream evaluations use linear probes trained on frozen representations.

(1) Cell-type composition regression: we predict the relative abundance of cell types in each patch using ridge regression. The target variable is a vector of cell-type proportions (excluding samples without annotations), and performance is reported via R^2.
(2) Niche-type classification: we classify patches into discrete tissue niche categories (for example, tumour, stroma or immune) using multinomial logistic regression. Performance is reported via F1 Macro.
(3) Niche consistency: a pairwise consistency task testing whether patches sharing the same niche label are embedded closer than patches from different niches. Positive pairs are patches with identical niche labels, while negative pairs are patches with different niche labels. We train a logistic regression probe on absolute embedding differences to predict whether a pair shares the same niche and report the area under the curve (AUC).
(4) Spatial consistency: a pairwise consistency task testing whether spatially nearby patches from the same slide are embedded closer than spatially distant patches. Positive pairs are patches within the k-nearest neighbours (default $k = 5$) on the same slide, while negative pairs are pairs beyond an 80th-percentile distance threshold on the same slide. We train a logistic regression probe on absolute embedding differences and report the AUC.
(5) Spatial neighbourhood prediction: we predict properties of neighbouring patches from centre-patch embeddings across increasing distance bins. For classification, we predict the majority niche label among neighbours from the centre embedding (via F1 Macro). For regression, we predict the mean cell-type composition of neighbours (via R^2). Performance is reported across distance bins to quantify degradation with increasing spatial separation.

Throughout, we treat linear-probe performance as a practical proxy for task-relevant information accessible from a given frozen representation. Cross-validation or predefined train/validation/test splits are used as appropriate for each task.

Class-wise thymus case study

For the thymus case study, we selected a representative held-out Visium slide and visualized class-wise niche prediction for GEX, imaging (IMG), Concat and CoMM (Extended Data Fig. 7 and Supplementary Section 8). For each selected niche class, we show ground-truth cells and correctly predicted cells for each method. We additionally report per-class F1 scores, row-normalized confusion matrices and class-wise SIS computed from per-class F1 relative to the stronger unimodal baseline. This analysis was used only as a qualitative interpretability case study and not for model selection.

Model training and experimental details

Frozen embeddings are extracted from pretrained foundation models: UNI-200M (UNI-2) for histopathology images[5] and Nicheformer for spatial transcriptomics[8]. Patches are sampled from full slides and paired at the patch level. We filter samples with `NaN` embeddings and handle missing annotations according to each task definition.

Training of the fusion-interface. For each multimodal method, we train the fusion mappings (projection heads) g_1 and g_2 on the training split using PyTorch Lightning. Methods with self-supervised fusion objectives (for example, CoMM, SimCLR, Barlow Twins, VICReg, BYOL and SimSiam) use all paired samples in the training split (without downstream labels), while evaluation always uses held-out data. We use early stopping (patience 50 epochs) based on a smoothed validation loss. For linear CCA, we fit the closed-form solution on the training split. All learned methods are trained with AdamW (learning rate and weight decay selected via a sweep) and cosine annealing learning-rate schedules (see Supplementary Information section 4 for more details).

Linear probe evaluation. After training the fusion interface, we train linear probes (logistic regression for classification; ridge regression for regression) on frozen representations (Z_1, Z_2 or Z_3). We use fivefold cross-validation or predefined train/validation/test splits depending on the evaluation setting. For spatial tasks, we enforce cross-slide validation to prevent spatial leakage across splits.

Probe-capacity control. To assess the dependence of SIS on readout capacity, we also computed a probe-indexed variant SIS_ϕ where we replace the linear probe used in the main analyses by shallow (`256-d`) or deep (`512-256-128`) MLP classifiers (ReLU, dropout set to 0.1, batch normalization; Adam lr = 10^{-3}, early stopping on the last 20% of each training fold; five random seeds; metrics averaged over folds and seeds). This analysis does not redefine the primary SIS metric, which is based on linear accessibility of task-relevant information but serves as a stress test for whether cross-modal information is present in a representation and requires a nonlinear decoder. We re-evaluate the frozen IMG, GEX, Concat and CoMM, and use the same readout as Table 1 on the same fivefold cross-validation splits and post-filter cohort as the main evaluation (NaN removal, excluded niche classes, rare-class filtering). At probe depth ϕ, we substitute Perf_ϕ into equation (15)

$$\widehat{\mathrm{SIS}}_\phi = \frac{\mathrm{Perf}_\phi(Z_3) - \max(\mathrm{Perf}_\phi(Z_1), \mathrm{Perf}_\phi(Z_2))}{\max(\mathrm{Perf}_\phi(Z_1), \mathrm{Perf}_\phi(Z_2))}. \quad (20)$$

In concatenated representations, an MLP probe implements supervised post hoc nonlinear fusion on fixed unimodal embeddings, $\hat{y} = h([Z_{\mathrm{IMG}}; Z_{\mathrm{GEX}}])$. This differs from DCCA-style objectives, which learn modality-specific nonlinear transformations through an unsupervised canonical-correlation objective. For CoMM, the representation is already learned by a synergy-aware nonlinear training objective; the MLP adds a second nonlinear readout stage that can make interaction information more accessible without changing the fusion objective itself (Extended Data Fig. 4 and Supplementary Information section 6).

Spectral analysis. Reference spectral subspaces are computed from N = 1,000 held-out test patches via the SVD of the centred cross-covariance $C = Z_1^{\mathrm{T}} Z_2 / N$. We use parallel analysis with L = 100 permutations to determine an effective dimensionality (typically K = 5). Canonical cosine between learned and reference subspaces is computed following standard numerical linear algebra protocols[54]; exact definitions are provided in Supplementary Information section 11.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The breast, lung and thymus spatial transcriptomics datasets used in this study are publicly available via their respective Hugging Face repositories at https://huggingface.co/datasets/theislab/CAMEO-Breast, https://huggingface.co/datasets/theislab/CAMEO-Lung and https://huggingface.co/datasets/theislab/CAMEO-Thymus. Pretrained model resources used for feature extraction include the Nicheformer GEX encoder[8], trained on the publicly available SpatialCorpus-110M dataset, and the UNI-2 IMG encoder[5], for which pretrained weights are publicly released while the underlying pretraining data are subject to institutional and regulatory constraints. For Nicheformer, we use the authors' released pretrained model checkpoint, provided via Mendeley Data at https://data.mendeley.com/preview/87gm9hrgm8?a=d95a6dde-e054-4245-a7eb-0522d6ea7dff. The SpatialCorpus-110M dataset is publicly available via Hugging Face at https://huggingface.co/datasets/theislab/SpatialCorpus-110M. For UNI-2, we use the publicly released `UNI2-h` model checkpoint available at https://huggingface.co/MahmoodLab/UNI2-h. Detailed descriptions of pretraining data sources and model checkpoints are provided in Supplementary Sections 2–3. Source data are provided with this paper.

Code availability

The code for computing SIS, spectral alignment analysis and all multimodal alignment methods evaluated in this study is publicly available via GitHub at https://github.com/microsoft/cell_synergy and via Zenodo at https://doi.org/10.5281/zenodo.22731603 (ref. 55). The repository includes implementations for all benchmarked methods, preprocessing pipelines and evaluation scripts to reproduce all results presented in this paper.

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Author contributions

T.R. and L.C. jointly designed the study and developed the theory and methods. T.R. and E.Z. implemented the method and performed the analyses. J.H. helped with developing theory. T.R., E.Z. and L.C. wrote the initial draft of the paper. F.J.T., S.R., P.S.W., A.P.A. and L.C. supervised the project. S.R., P.S.W., A.P.A. and L.C. provided resources. All authors contributed to interpreting the results and revising the paper.

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Competing interests

F.J.T. consults for Immunai, CytoReason, BioTuring, Genbio and Valinor Industries, and has ownership interest in RN.AI Therapeutics, Dermagnostix and Cellarity. S.R. holds equity in Amgen. S.R. and P.S.W. received research funding from Microsoft. E.Z., J.H., A.P.A. and L.C. are employees of Microsoft and own equity in Microsoft. P.S.W. reports compensation for consulting/speaking from Engine Ventures, AbbVie and Passkey Therapeutics unrelated to this work. The other authors declare no competing interests.

Additional information

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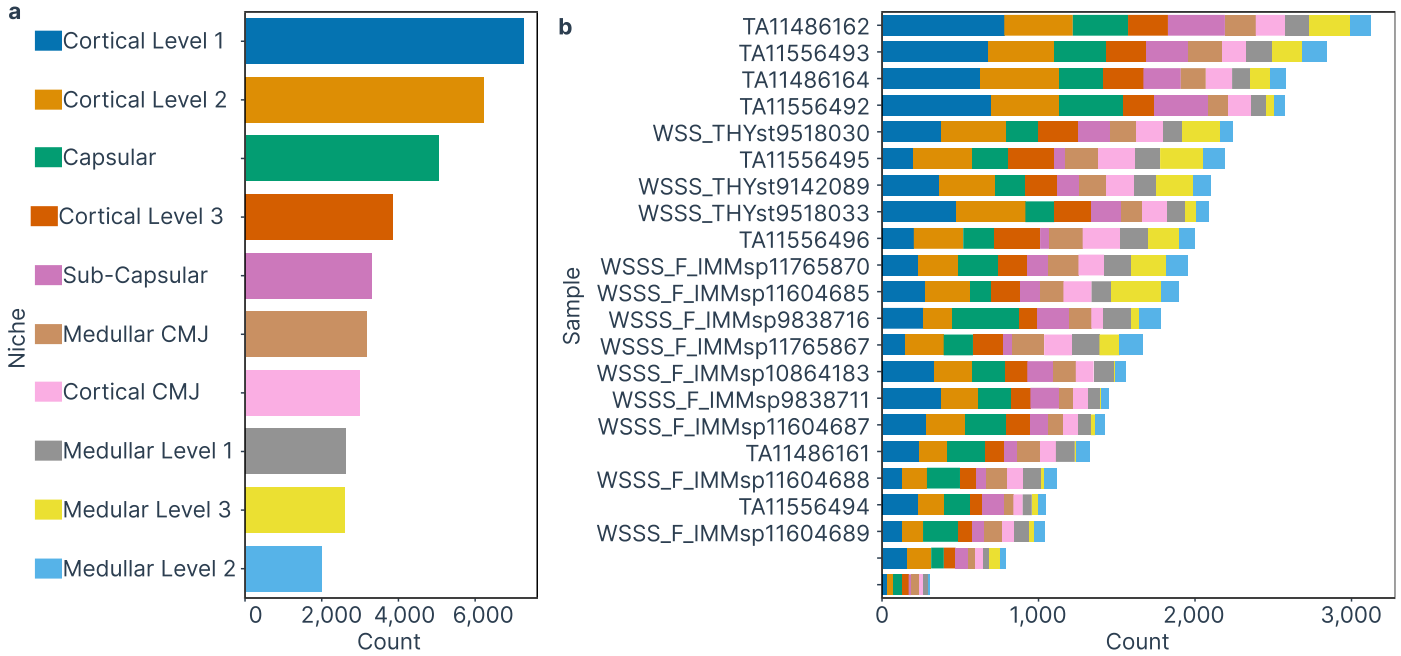
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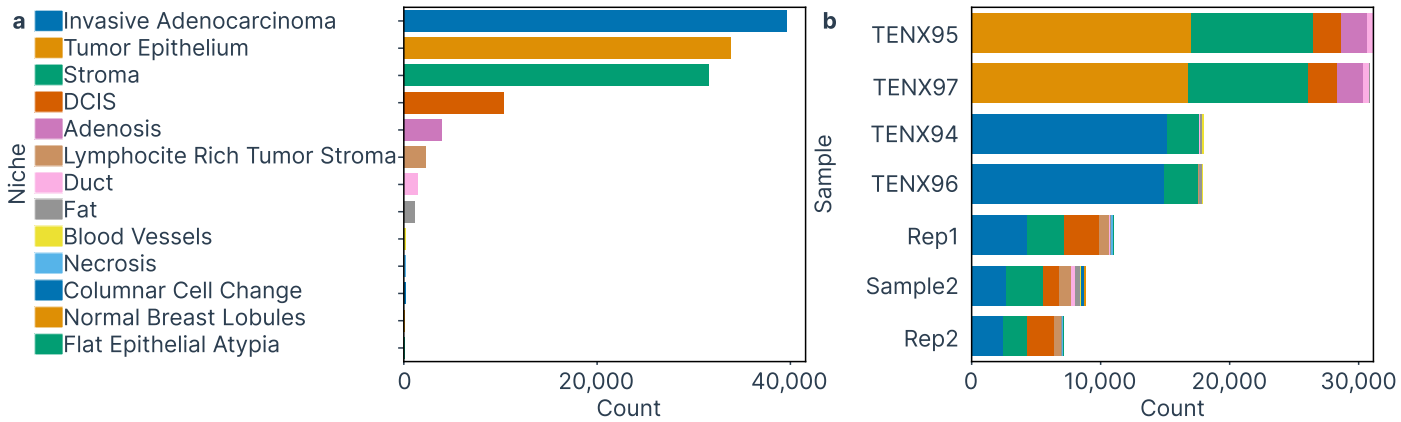
Extended Data Fig. 1 | Data exploration for the lung dataset. a, Overall distribution of histopathological niches, excluding *NOANNOT* patches without niche annotation. The lung dataset exhibits a characteristic fibrotic pattern with structural niches (alveolar regions, bronchioles) and pathological features (fibrosis, inflammation). **b**, Niche distribution per sample shows heterogeneity across patients (2-13 distinct niches per sample). **c**, Overall cell-type composition showing the dominance of structural cells (fibroblasts, epithelial), with immune infiltration (macrophages, lymphocytes). **d**, Per-sample cell-type composition reveals patient-specific immune responses and fibrotic progression patterns. The *NOMAP* category represents cells excluded from evaluation.

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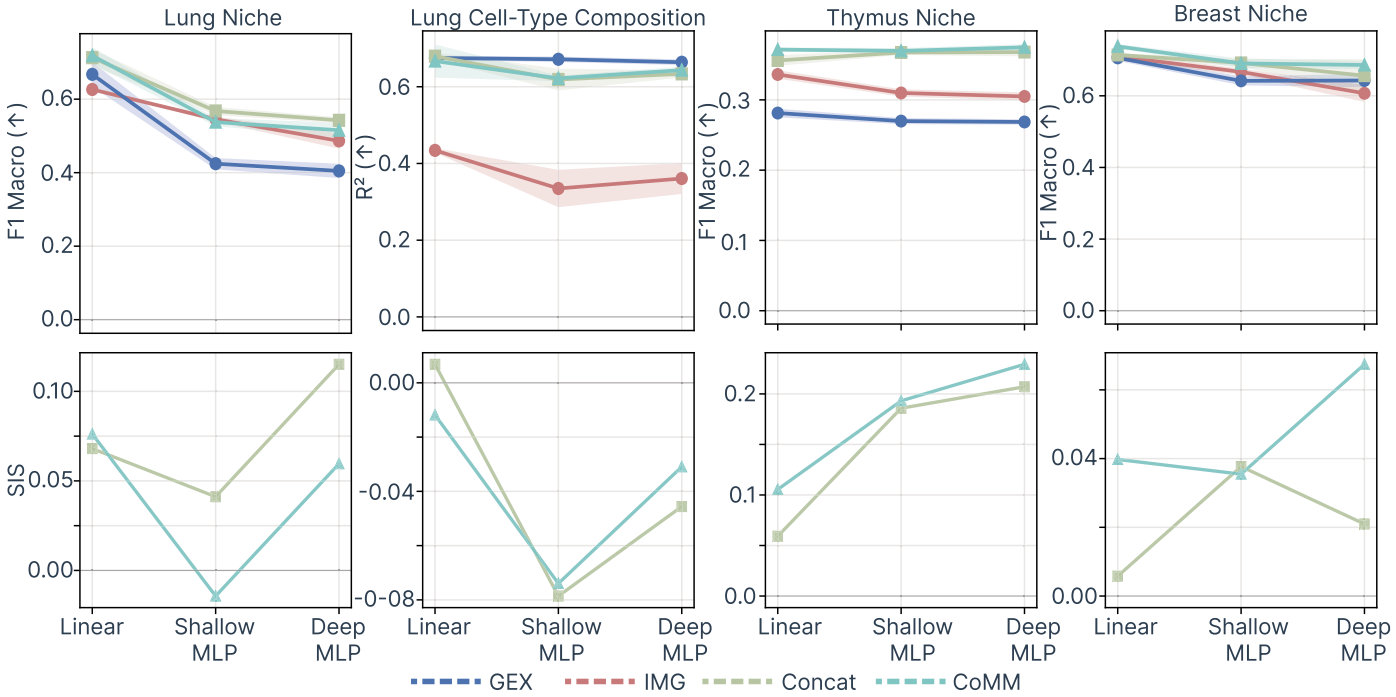
Extended Data Fig. 2 | Data exploration for the thymus dataset. a, Overall niche distribution excluding unassigned regions. The thymus exhibits highly structured developmental niches (cortex, medulla, cortico-medullary junction) reflecting T-cell maturation zones. **b**, Niche distribution per sample demonstrates consistent developmental architecture across fetal and postnatal stages (10-11 distinct niches per sample, highest heterogeneity among all datasets). No cell-type composition labels are available for this spot-based Visium dataset.

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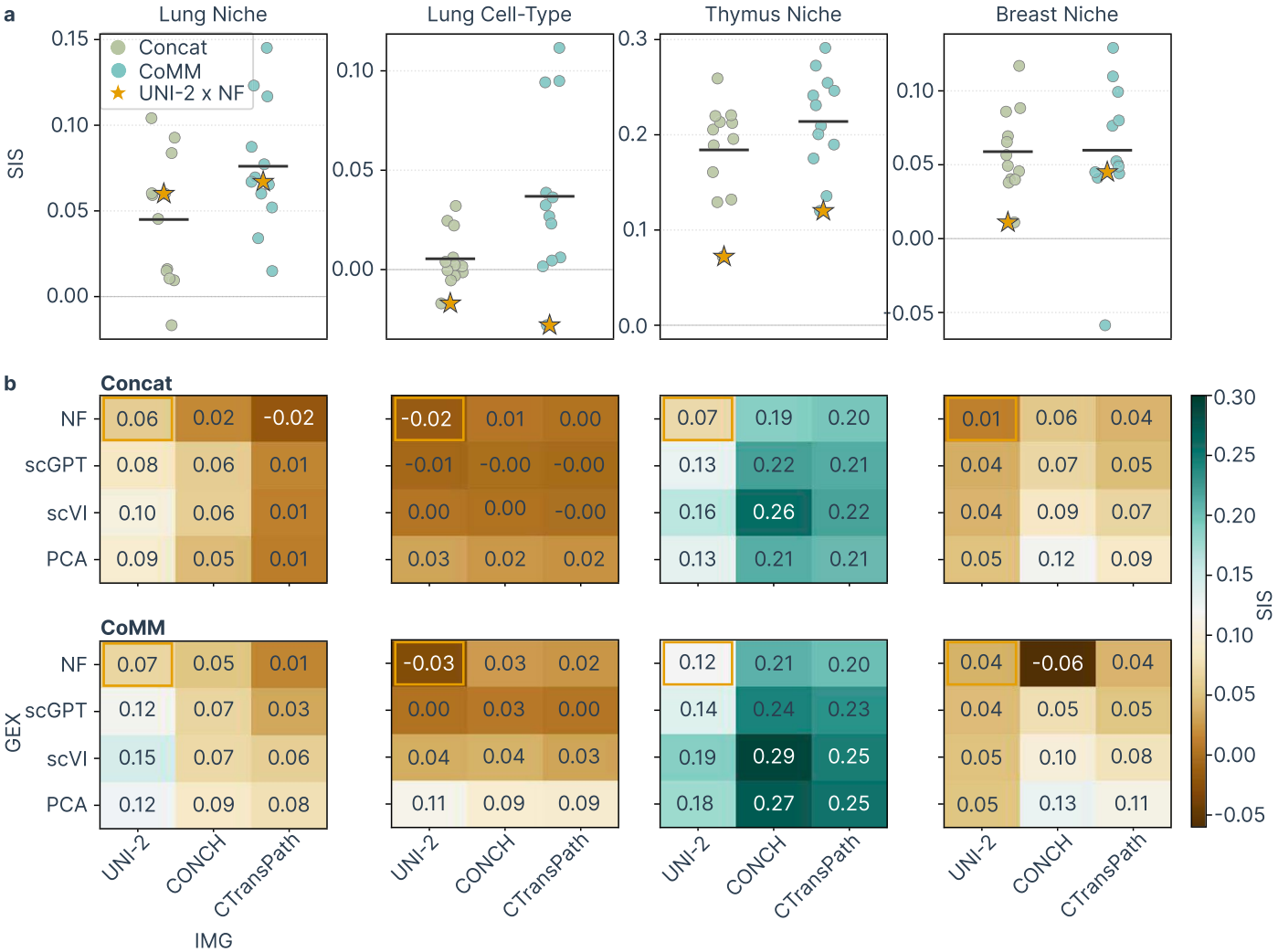
Extended Data Fig. 3 | Data exploration for the breast dataset. a, Overall niche distribution excluding *NOANNOT* patches. The breast cancer samples show tumor microenvironment complexity with distinct tumor regions, stromal compartments, immune infiltration zones, and adipose tissue. **b**, Niche distribution per sample reveals tumor heterogeneity (7-10 distinct niches per sample), with variable immune and stromal content reflecting different cancer subtypes and stages. No cell-type composition labels are available for certain samples (marked with "*" in Supplementary Table 1).

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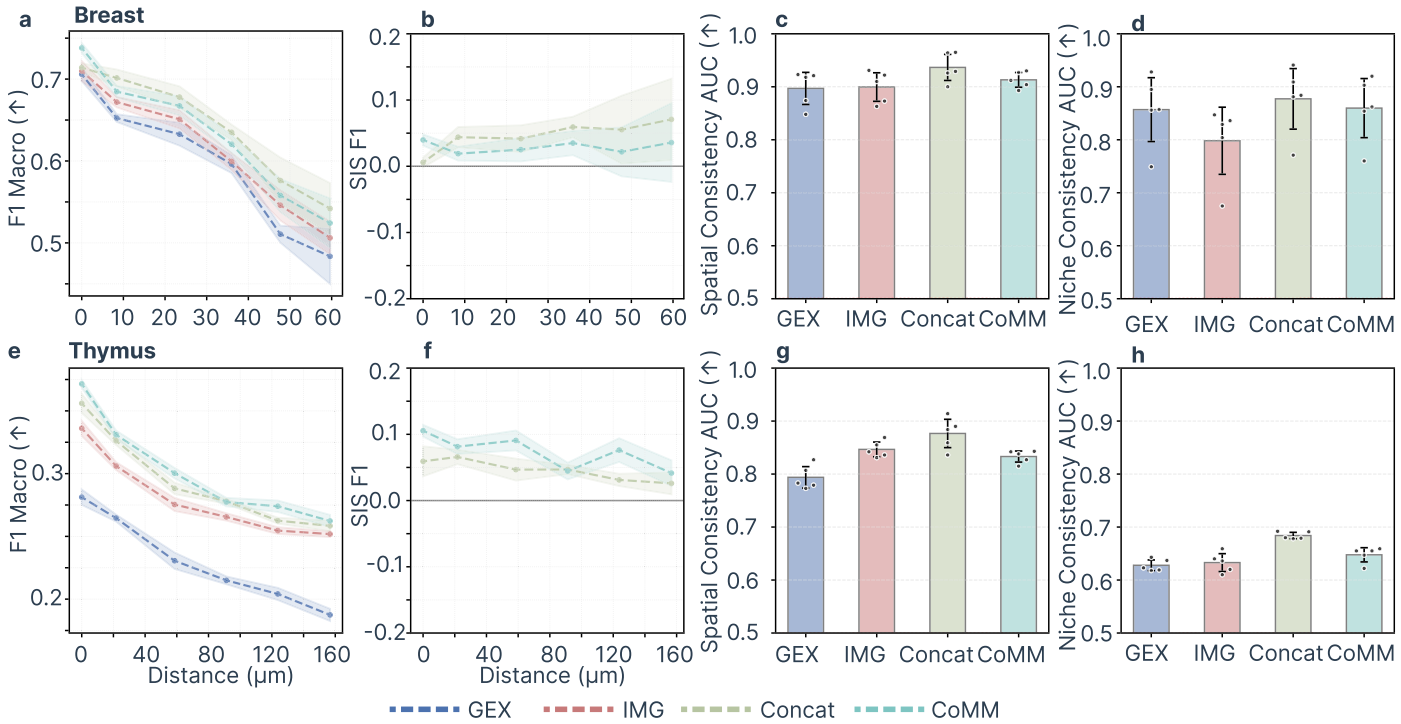
Extended Data Fig. 4 | Probe-capacity control: task performance and SISϕ. **Top:** Macro F1 (niche) or R^2 (lung cell-type composition) for IMG, GEX, Concat, and CoMM (`concat_aug_l2`) under linear, shallow (`256-d`), and deep (`512-256-128`) MLP probes. Shaded bands show ±1 standard deviation over five CV-folds and five MLP random seeds. **Bottom:** SIS_φ for Concat and CoMM only, using the same unimodal baselines at each probe depth. **Tasks across columns from left to right:** Lung niche, lung cell-type composition, thymus niche, and breast niche.

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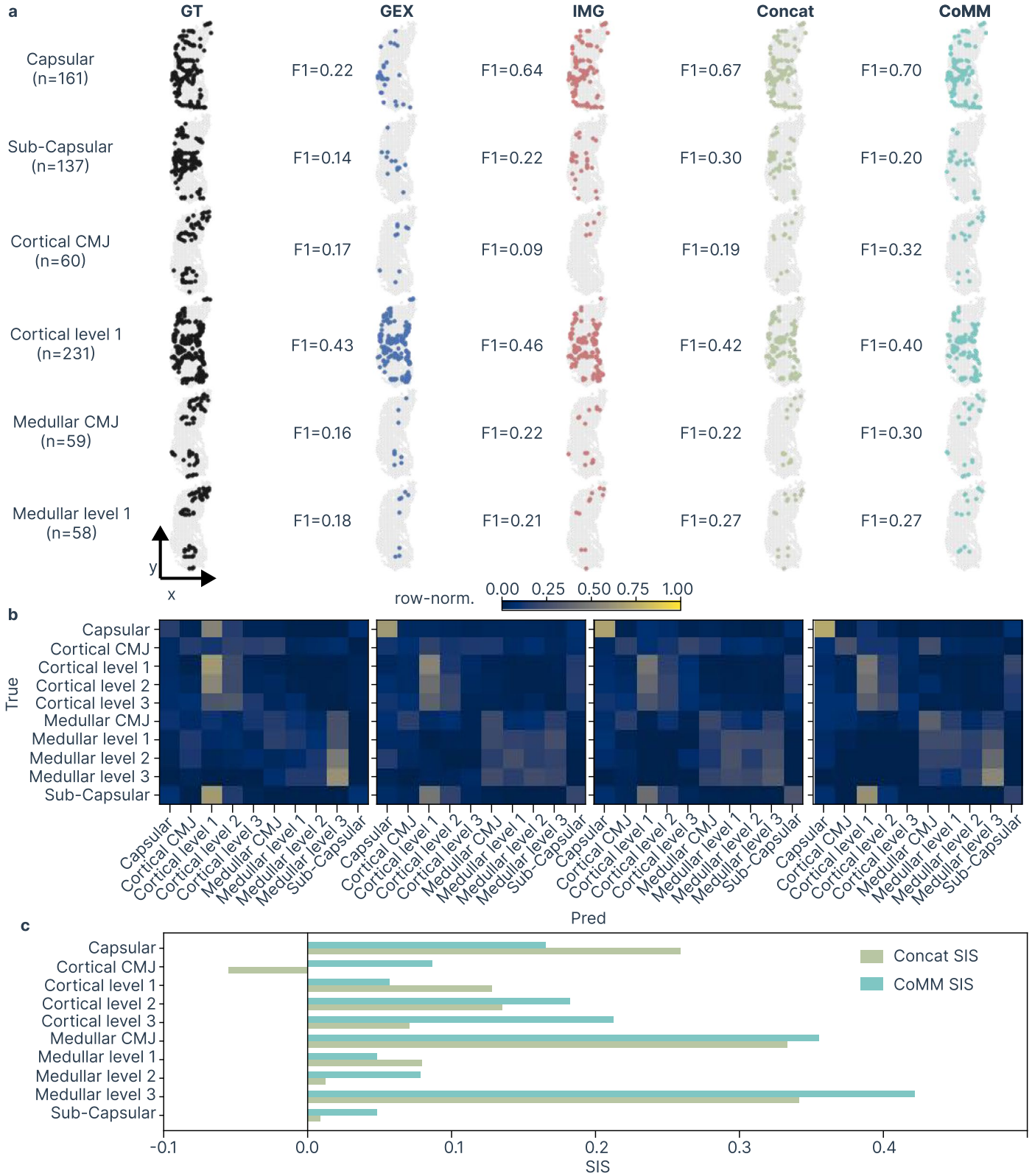
Extended Data Fig. 5 | Assessing the sensitivity of SIS to the choice of encoder-backbone pair. a, Distribution of SIS values across 12 encoder pairs using three image encoders (UNI-2, CONCH, CTransPath) and four gene expression representations (Nicheformer, scGPT, scVI, PCA). Points denote encoder pairs, horizontal bars denote means, and orange stars mark the default UNI-2 × Nicheformer (NF) pair used in the main benchmark. **b**, SIS heatmaps for Concat and CoMM across the same encoder grid. Orange outlines mark the default UNI-2 × Nicheformer pair used in the main benchmark. SIS is computed with encoder-matched unimodal baselines and the same five-fold linear probe protocol as the main experiments. The analysis tests robustness of SIS regimes to backbone choice and is not intended as a benchmark of the external encoders.

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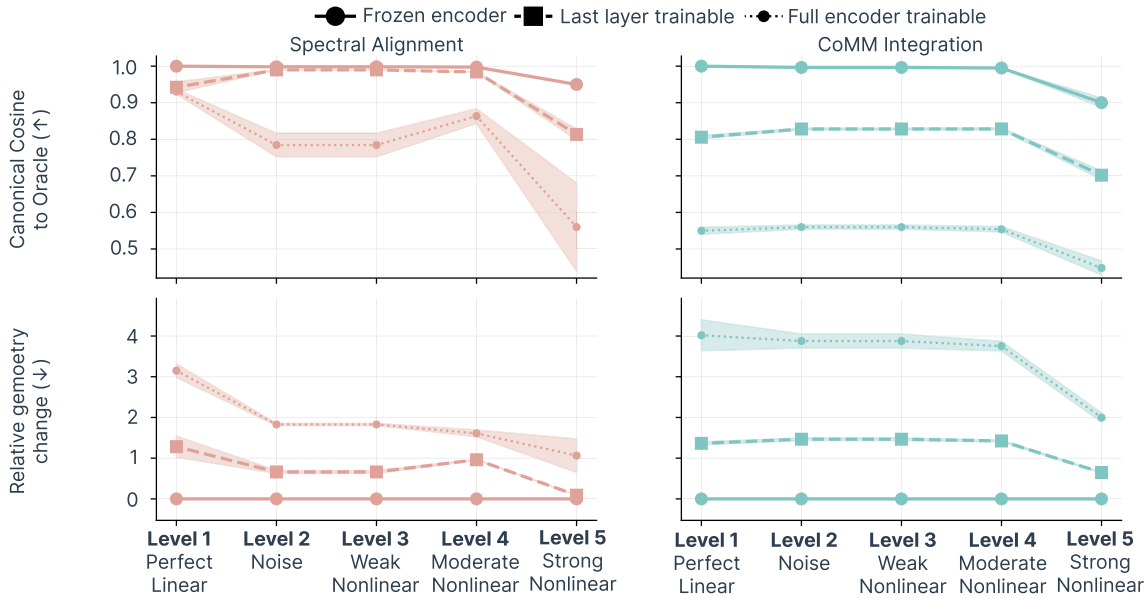
Extended Data Fig. 6 | Extended results for the breast and thymus datasets. Breast (redundancy-dominated; panels **a**-**d**): **a**, Macro F1 for neighbor niche prediction across distance bins; **b**, SIS computed from F1; **c**, Spatial consistency (area under the curve; AUC); **d**, Niche consistency (AUC). **Thymus (resolution-mismatched; panels e-h):e**, Macro F1 for neighbor niche prediction; **f**, SIS computed from F1; **g**, Spatial consistency (AUC); **h**, Niche consistency (AUC). Shaded regions and error bars denote variability across folds.

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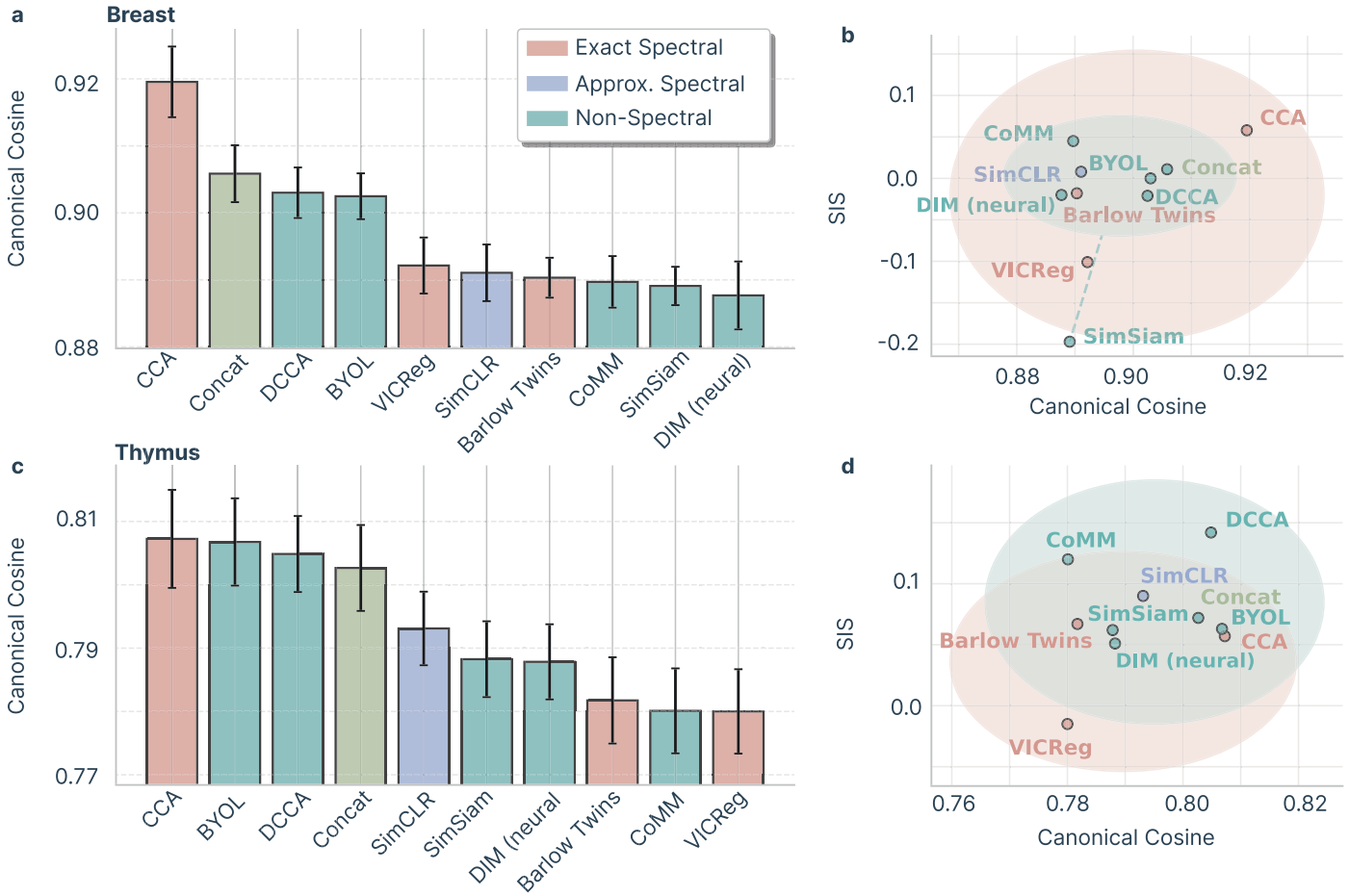
Extended Data Fig. 7 | Case study results assessing thymus niche recovery on section TA11486161. a, Ground-truth patches and correctly predicted patches for six representative thymus niches across GEX, IMG, Concat, and CoMM. Shown are the per-class macro F1 for each method. **b**, Row-normalized confusion matrices for the same section. **c**, Dataset-wide per-niche SIS for Concat and CoMM, computed against the best unimodal baseline using identical out-of-fold linear probes. Throughout the figure: IMG = imaging, GEX = gene expression, and Concat = Concatenation.

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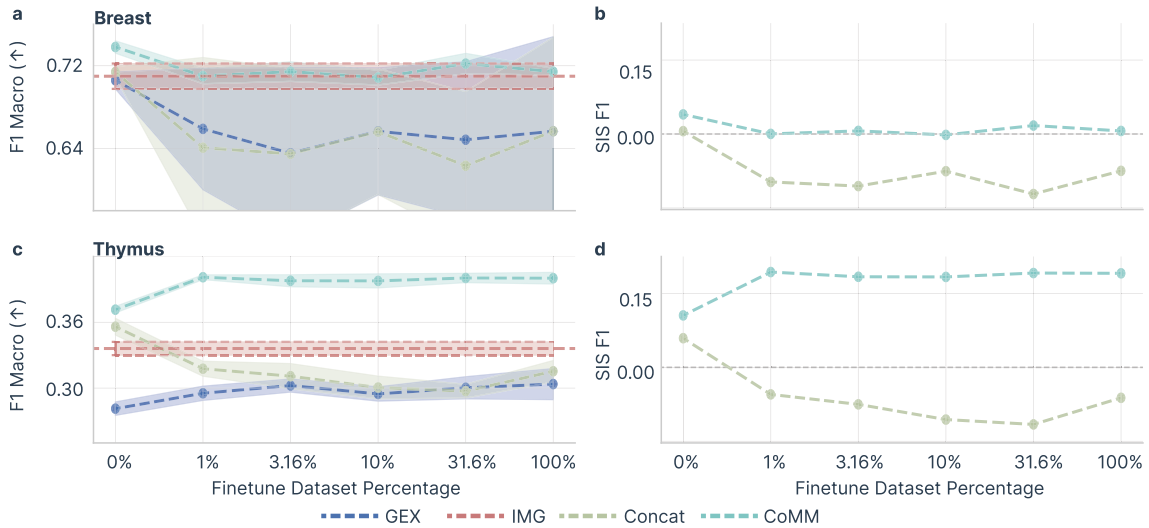
Extended Data Fig. 8 | Encoder trainability changes the geometry underlying the spectral ceiling. We use the same five-level synthetic ladder as Fig. 3, ranging from linear shared structure to strong nonlinear coupling. **Columns:** spectral-alignment objective (left) and CoMM-style integration objective (right). **Top:** canonical cosine to the oracle shared generative directions, measuring recovery of the true shared signal. **Bottom:** relative change in cross-modal representation geometry from initialization (note that the frozen encoders remain at zero by construction). Line styles indicate encoder trainability: solid = frozen encoder, dashed = last layer trainable, and dotted = full encoder trainable. Shaded bands denote mean ± SD over seeds ($n = 5$).

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Extended Data Fig. 9 | Spectral ceiling and synergy across datasets. Breast (top):a, canonical cosine quantifying convergence to the linear subspace recovered by spectral solutions; **b**, alignment-synergy landscape (SIS versus canonical cosine). **Thymus (bottom):c**, canonical cosine; **d**, SIS versus canonical cosine. Error bars and shaded regions denote variability across five folds.

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Extended Data Fig. 10 | Extended scaling analysis. Breast (top): **a**, Macro F1 for niche classification versus fraction of data used for fine-tuning; and **b**, corresponding SIS as a function of the fraction of data used for fine-tuning. **Thymus (bottom):c**, Macro F1 versus fraction of fine-tuning fraction; and **d**, corresponding SIS.

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Policy information about availability of computer code

Data collection

We evaluated all fusion methods on three human spatial transcriptomics datasets to validate robustness and assess how dataset characteristics influence the utility of multimodal integration. The dataset was obtained on 01 July 2026 from the publicly available HuggingFace repository (https://huggingface.co/datasets/theislab/CAMEO-Lung; https://huggingface.co/datasets/theislab/CAMEO-Thymus; https://huggingface.co/datasets/theislab/CAMEO-Breast). Briefly, the authors integrated spatial transcriptomics measurements (10x Xenium and Visium) with matched histopathology image patches and curated expert-defined niche annotations across three tissues (lung, breast, and thymus). This yielded a partially paired multimodal dataset with aligned image and gene expression embeddings and tissue-specific ground-truth region labels. These datasets differ systematically in size, spatial resolution, data acquisition technology, and label availability (Supplementary Table 1), enabling us to test our theoretical predictions across distinct information regimes.

The lung and breast datasets utilize Xenium in situ technology, offering subcellular resolution (approximately 0.21 µm/pixel) with precise spatial correspondence between histopathology and gene expression. In contrast, the thymus dataset uses Visium spatial transcriptomics, which aggregates expression within 55 µm spots (capturing typically 1-10 cells per spot). This fundamental difference in spatial resolution creates distinct integration challenges: Xenium data presents a redundancy-dominated regime where morphology and expression are tightly aligned, while Visium data presents a resolution-mismatched regime requiring learned integration to bridge the scale gap between cellular histology and multi-cellular expression aggregates.

Data analysis

The code for computing the Synergistic Information Score (SIS), spectral alignment analysis, and all multimodal alignment methods evaluated in this study is publicly available at https://github.com/microsoft/cell_synergy. The repository includes implementations for all benchmarked methods, preprocessing pipelines, and evaluation scripts to reproduce all results presented in this manuscript.

We use the pretrained Nicheformer encoder described in Tejada-Lapuerta et al. Nicheformer is pretrained on SpatialCorpus-110M, a large-

scale collection comprising approximately 57.06 million dissociated single-cell transcriptomes and 53.83 million image-based spatially resolved cells across 73 human and mouse tissues. The SpatialCorpus-110M dataset is publicly available via Hugging Face at https://huggingface.co/datasets/theislab/SpatialCorpus-110M. We use the authors' released pretrained model checkpoint, provided through Mendeley Data at https://data.mendeley.com/preview/87gm9hrgm8?a=d95a6dde-e054-4245-a7eb-0522d6ea7dff. Unless explicitly stated otherwise (e.g., for fine-tuning experiments), Nicheformer is treated as a frozen encoder.

We use UNI-2 (UNI2-h) as described in Chen et al. UNI-2 is pretrained using self-supervised learning on a large corpus of histopathology image tiles (>200M tiles) derived from over 300k-350k whole-slide images sourced from internal clinical archives at Mass General Brigham. As detailed in the corresponding publication, the underlying pretraining data are not fully publicly released due to institutional and regulatory constraints, and access to curated datasets is evaluated on a case-by-case basis. We use the publicly released UNI2-h model checkpoint available at https://huggingface.co/MahmoodLab/UNI2-h. Unless explicitly noted, UNI-2 is used as a frozen image encoder.

In the main text, we compare the performance of different multimodal fusion methods. All descriptions for these approaches can be found in Supplementary Section 3.

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The breast, lung, and thymus spatial transcriptomics datasets used in this study are publicly available from their respective Hugging Face repositories: https://huggingface.co/datasets/theislab/CAMEO-Breast, https://huggingface.co/datasets/theislab/CAMEO-Lung, and https://huggingface.co/datasets/theislab/CAMEO-Thymus. Pretrained model resources used for feature extraction include the Nicheformer gene expression (GEX) encoder, trained on the publicly available SpatialCorpus-110M dataset, and the UNI-2 imaging (IMG) encoder, for which pretrained weights are publicly released while the underlying pretraining data are subject to institutional and regulatory constraints. For Nicheformer, we use the authors' released pretrained model checkpoint, provided through Mendeley Data at https://data.mendeley.com/preview/87gm9hrgm8?a=d95a6dde-e054-4245-a7eb-0522d6ea7dff. The SpatialCorpus-110M dataset is publicly available via Hugging Face at https://huggingface.co/datasets/theislab/SpatialCorpus-110M. For UNI-2, we use the publicly released UNI2-h model checkpoint available at https://huggingface.co/MahmoodLab/UNI2-h. Detailed descriptions of pretraining data sources and model checkpoints are provided in the Supplementary Sections 2-3.

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